CSSMedia CSS Reference



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CSS Reference

Look up every selector, property, value and at-rule by subject.

This book is a complete reference for CSSMedia. Each table links to the CSS specification that defines its entries. The references are grouped by subject:

- Selector reference
- Cascade reference
- Margins, padding and borders
- Box sizes
- Backgrounds
- Shadows, opacity, transforms and masks
- Display and positioning
- Tables
- Text and fonts
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- Images
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- MathML

Selector reference

These tables list every selector that CSSMedia accepts. The `In a PDF` column says whether the selector matches elements while CSSMedia generates a PDF.

Basic selectors

Basic selectors match an element by its name, ID or attributes. See Basic selectors and namespaces.

Name	Description	Values	In a PDF	Specification
`E`	Type selector. Matches elements named `E`.	An element name.	Matches	Selectors 4 §5.1
`*`	Universal selector. Matches any element.		Matches	Selectors 4 §5.2
`#id`	ID selector. Matches the element whose `id` or `xml:id` attribute has this value.	An identifier.	Matches	Selectors 4 §6.7
`.name`	Class selector. Matches inside SVG and MathML content, which define a `class` attribute.	An identifier.	Matches in SVG and MathML only	Selectors 4 §6.6
`[attr]`	Matches elements that have the attribute.	An attribute name, with an optional namespace prefix.	Matches	Selectors 4 §6
`[attr=value]`	Matches when the attribute value is exactly `value`.	A string or an identifier. Add `i` to ignore ASCII case or `s` to require it.	Matches	Selectors 4 §6.1
`[attr~=value]`	Matches when the value is one of the white-space separated words of the attribute.	A word without white space.	Matches	Selectors 4 §6.1
`[attr|=value]`	Matches when the attribute is `value` or starts with `value-`.	A string or an identifier.	Matches	Selectors 4 §6.1

Name	Description	Values	In a PDF	Specification
`[attr^=value]`	Matches when the attribute starts with `value`.	A non-empty string.	Matches	Selectors 4 §6.2
`[attr$=value]`	Matches when the attribute ends with `value`.	A non-empty string.	Matches	Selectors 4 §6.2
`[attr*=value]`	Matches when the attribute contains `value`.	A non-empty string.	Matches	Selectors 4 §6.2
`ns|E`, `*|E`, `|E`	Limits a type selector to a declared namespace, to any namespace, or to no namespace. The same prefixes work in attribute selectors.	A prefix declared with `@namespace`.	Matches	Selectors 4 §5.3, CSS Namespaces 3 §3
`&`	Nesting selector. Stands for the selector of the parent rule.		Matches	CSS Nesting 1

Combinators

A combinator relates two selectors. See Combinators.

Name	Description	Values	In a PDF	Specification
`A B`	Descendant combinator. Matches `B` inside `A` at any depth.	Two selectors separated by white space.	Matches	Selectors 4 §14.1
`A > B`	Child combinator. Matches `B` when its parent is `A`.	Two selectors.	Matches	Selectors 4 §14.2
`A + B`	Next-sibling combinator. Matches `B` when it directly follows `A`.	Two selectors.	Matches	Selectors 4 §14.3
`A ~ B`	Subsequent-sibling combinator. Matches `B` when `A` is an earlier sibling.	Two selectors.	Matches	Selectors 4 §14.4
`A || B`	Column combinator. Matches the cells `B` of the column that `A` describes.	Two selectors.	Matches	Selectors 5 §8.1

Name	Description	Values	In a PDF	Specification
`A / attr/ B`	Reference combinator. Matches `B` when the attribute `attr` of `A` refers to its ID.	Two selectors and an attribute name between slashes.	Matches	Selectors 5 §9.1

Pseudo-classes

A pseudo-class tests a property of an element other than its name and attributes. See Pseudo-classes.

Name	Description	Values	In a PDF	Specification
`:root`	The root element of the document.		Matches	Selectors 4 §13.1
`:empty`	An element with no child elements and no text other than white space.		Matches	Selectors 4 §13.2
`:scope`	The scoping element. Without another scope, the root element.		Matches	Selectors 4 §8.4
`:first-child`, `:last-child`, `:only-child`	The first, last or only element among its siblings.		Matches	Selectors 4 §13.3
`:first-of-type`, `:last-of-type`, `:only-of-type`	The first, last or only element of its type among its siblings.		Matches	Selectors 4 §13.4
`:nth-child()`, `:nth-last-child()`	Elements at a position among their siblings, counted from the start or from the end.	odd, even or `An+B`, optionally followed by `of` and a selector list.	Matches	Selectors 4 §13.3
`:nth-of-type()`, `:nth-last-of-type()`	Elements at a position among the siblings of the same type.	odd, even or `An+B`.	Matches	Selectors 4 §13.4

Name	Description	Values	In a PDF	Specification
`:nth-col()`, `:nth-last-col()`	Table cells at a position among the columns of their table.	`odd`, `even` or `An+B`.	Matches	Selectors 5 §8.2, §8.3
`:is()`, `:matches()`	An element that matches any selector of the list. `:matches()` is the old name of `:is()`.	A selector list. Invalid selectors are ignored.	Matches	Selectors 4 §4.2
`:where()`	Works as `:is()` and adds no specificity.	A selector list. Invalid selectors are ignored.	Matches	Selectors 4 §4.4
`:not()`	An element that matches none of the selectors of the list.	A selector list. An invalid selector makes the whole selector invalid.	Matches	Selectors 4 §4.3
`:has()`	An element that has a related element matching the selector.	A relative selector list, such as `> figure`. It cannot contain another `:has()` or a pseudo-element.	Matches	Selectors 4 §4.5
`:lang()`	Elements in a language. The language comes from `xml:lang` or the `-cssmedia-lang` property.	One or more language ranges, as identifiers or strings.	Matches	Selectors 4 §7.2
`:dir()`	Elements by text direction, read from the text and the language, not from the `direction` property.	`ltr` or `rtl`.	Matches	Selectors 4 §7.1

Name	Description	Values	In a PDF	Specification
`:link`, `:any-link`	Elements that are hyperlinks, because `-cssmedia-link` gives them a destination.		Matches	Selectors 4 §8.1, §8.2
`:local-link`	Hyperlinks that point to the same document. The optional number limits how many path segments are compared.	An optional non-negative integer in parentheses.	Matches	Selectors 5 §2.1
`:read-only`, `:defined`	Every element is read-only and defined, because an XML element has no editable or undefined state.		Always matches	Selectors 4 §12.1.2, §5.4
`:hover`, `:active`, `:focus`, `:focus-visible`, `:focus-within`	User action states, such as pointing at an element or giving it keyboard focus.		Never matches	Selectors 4 §9
`:interest-source`, `:interest-target`	Source and target of a user's declared interest.		Never matches	Selectors 5 §3.1
`:visited`, `:target`	Links that were visited, and the element that a link points to. A PDF has no browsing history and no current target.		Never matches	Selectors 4 §8.2, §8.3
`:playing`, `:paused`, `:seeking`, `:buffering`, `:stalled`, `:muted`, `:volume-locked`	Media playback, loading and sound states.		Never matches	Selectors 4 §10

Name	Description	Values	In a PDF	Specification
`:current`, `:current()`, `:past`, `:future`	Elements in relation to a timeline, such as speech or video playback.	A selector list for `:current()`.	Never matches	Selectors 5 §5
`:open`, `:popover-open`, `:modal`, `:fullscreen`, `:picture-in-picture`	Display states of interactive elements.		Never matches	Selectors 4 §11
`:enabled`, `:disabled`, `:read-write`, `:placeholder-shown`, `:autofill`, `:default`, `:checked`, `:unchecked`, `:indeterminate`, `:blank`	Form control states. An XML element is never a form control.		Never matches	Selectors 4 §12, Selectors 5 §4.1
`:valid`, `:invalid`, `:in-range`, `:out-of-range`, `:required`, `:optional`, `:user-valid`, `:user-invalid`	Form validation states.		Never matches	Selectors 4 §12.3
`:state()`	A custom state that a document language exposes. XML has none.	An identifier.	Never matches	Selectors 5 §6
`:heading`, `:heading()`	Headings, as a document language defines them. CSSMedia does not give XML elements a heading role.	A list of levels for `:heading()`.	Never matches	Selectors 5 §7

Pseudo-elements

A pseudo-element selects a part of an element or generated content. See Pseudo-elements.

Name	Description	Values	In a PDF	Specification
`::before`, `::after`	Generated content before or after the content of an element. The `content` property gives it.		Matches	CSS Pseudo-Elements 4 §4.1
`::marker`	The marker of a list item. Also `::before::marker` and `::after::marker`.		Matches	CSS Pseudo-Elements 4 §4.2
`::first-line`	The first formatted line of a block.		Matches	CSS Pseudo-Elements 4 §2.1
`::first-letter`	The first letter of a block, with the punctuation around it. `::first-letter::prefix` and `::first-letter::suffix` select that punctuation.		Matches	CSS Pseudo-Elements 4 §2.2
`::footnote-call`	The reference that replaces a footnote in the text. By default it is the value of the `footnote` counter, as a superscript.		Matches	CSS GCPM 3 §2.6
`::footnote-marker`	The marker at the start of a footnote body. By default it is the value of the `footnote` counter, followed by a period and a space.		Matches	CSS GCPM 3 §2.7

Name	Description	Values	In a PDF	Specification
`::selection`, `::search-text`, `::target-text`, `::spelling-error`, `::grammar-error`, `::highlight()`	Highlighted ranges created by selection, search, navigation, spelling and grammar checking, or scripts.	A custom identifier for `::highlight()`.	No range exists	CSS Pseudo-Elements 4 §3.1
`::placeholder`, `::file-selector-button`, `::details-content`	Parts of form controls and disclosure widgets.		No such part exists	CSS Pseudo-Elements 4 §4.3, §5.1, §5.2
`::backdrop`	The backdrop of an element shown in the top layer.		No such part exists	CSS Position 4 §3.2

Cascade reference

The `Default` column lists values used by CSSMedia's current PDF-oriented rule evaluation. Media-query width, height and ratio values use the A4 default page size.

Property and at-rule reference

Name	Description	Values	Default	Specification
`all`	Resets all properties except `direction` and `unicode-bidi`.	`initial` | `inherit` | `unset` | `revert` | `revert-layer`		CSS Cascade 5 §3.2
`@charset`	Declares the stylesheet's fallback character encoding.	`@charset "<label>";`		CSS Syntax 3 §8.3

Name	Description	Values	Default	Specification
`@import`	Loads another stylesheet, optionally conditioned on a layer, support test or media query.	`@import <url-or-string> [layer | layer(<layer-name>)]? [supports(<condition>)]? <media-query-list>?;`		CSS Cascade 5 §2
`@namespace`	Declares a namespace prefix for selectors.	`@namespace <namespace-prefix>? <string-or-url>;`		CSS Namespaces 3 §3
`@layer`	Declares layer order or groups rules in a cascade layer.	`@layer <layer-name>? { <stylesheet> }` or `@layer <layer-name-list>;`		CSS Cascade 5 §6.4
`@property`	Registers a custom property with syntax, inheritance and optional initial value.	`@property <custom-property-name> { <declaration-list> }`		CSS Properties and Values API 1 §3
`syntax`	Defines the grammar of a registered custom property.	`<string>`		CSS Properties and Values API 1 §3.1
`inherits`	Sets whether a registered custom property inherits by default.	`true` | `false`		CSS Properties and Values API 1 §3.2
`initial-value`	Sets the registered custom property's initial value.	`<declaration-value>`	Guaranteed-invalid for `syntax: "*";` otherwise required	CSS Properties and Values API 1 §3.3

Name	Description	Values	Default	Specification
`@media`	Includes rules when a media query matches the PDF environment.	`@media <media-query-list> { <rule-list> }`		CSS Conditional 3 §5
`@supports`	Includes rules when CSSMedia accepts a declaration, selector or supported feature test.	`@supports <supports-condition> { <rule-list> }`		CSS Conditional 3 §6
`@supports-condition`	Defines a named group of support tests.	`@supports-condition <supports-condition-name> { <block-contents> }`		CSS Conditional 5 §8
`@container`	Includes rules when a selected ancestor container passes the query.	`@container <container-condition># { <block-contents> }`		CSS Conditional 5 §5.4
`@scope`	Limits the selectors in a rule list to a scope root and optional scope limit.	`@scope [(<scope-start>)]? [to (<scope-end>)]? { <rule-list> }`		CSS Cascade 6 §2.5
`@custom-media`	Gives a name to a media query list or boolean value.	`@custom-media <extension-name> [<media-query-list> | true | false];`		Media Queries 5 §10
`@when`	Includes rules when a generalized boolean condition is true.	`@when <boolean-condition> { <rule-list> }`		CSS Conditional 5 §3

Name	Description	Values	Default	Specification
`@else`	Supplies the next branch of a conditional chain.	`@else <boolean-condition>? { <rule-list> }`		CSS Conditional 5 §4
`@function`	Defines a custom CSS function.	`@function <name>(<parameters>) [returns <type>]? { <declaration-rule-list> }`		CSS Functions and Mixins 1 §2.1
`result`	Supplies a custom function's return value.	`<declaration-value>?`		CSS Functions and Mixins 1 §2.2

- `?` means optional, `|` means choose one, `#` means one or more comma-separated items, `||` means one or more options in any order, and brackets group parts of the syntax.
- `<label>` is an encoding name.
- `<url-or-string>` and `<string-or-url>` each mean a URL or quoted string.
- `<namespace-prefix>` is the prefix bound to a namespace.
- `<layer-name>` names a cascade layer. `<layer-name-list>` is a comma-separated list of layer names.
- `<stylesheet>` is a CSS stylesheet.
- `<custom-property-name>` is a custom property name beginning with `--`.
- `<declaration-list>` is a list of declarations.
- `<block-contents>` is content inside a rule's braces, including declarations and nested rules.
- `<declaration-value>` is the value part of a declaration: one or more CSS tokens, such as `navy`, `calc(1em + 2px)` or `var(--size)`. It cannot contain a top-level semicolon or `!`, since those delimit declarations and `!important`. Semicolons and exclamation marks inside functions are not top-level. See CSS Syntax 3.
- `<string>` is a quoted CSS string.
- `<media-query-list>` is one or more comma-separated media queries.
- `<rule-list>` is a sequence of rules.
- `<condition>` and `<supports-condition>` are support conditions made from feature tests and boolean operators.
- `<supports-condition-name>` names a reusable support condition.
- `<container-condition>` optionally names a query container and gives its query.
- `<boolean-condition>` combines `media()` and `supports()` tests.
- `<scope-start>` and `<scope-end>` are selector lists that set the scope roots and limits of `@scope`.

- `<extension-name>` names a custom media query.
- `<name>` in the function syntax is a function name beginning with `--`.
- `<parameters>` is a comma-separated list of function parameters.
- `<type>` is the optional syntax expected for the function result.
- `<declaration-rule-list>` contains declarations and nested rules.
- `<number>` in the function example is a CSS number, which can be an integer or include a fractional part.

The `@charset` rule must be the first rule in a stylesheet to affect byte decoding. Use `@namespace` before ordinary rules, and place `@import` rules before namespace declarations and ordinary rules.

Function names start with `--`. Parameters can have a syntax type and a default value. Use the `result` descriptor for the returned value:

```
@function --double(--value <number>) returns <number> {
    result: calc(var(--value) * 2);
}
```

`@supports` declaration tests use parentheses. `selector()` tests one complex selector. `font-tech()`, `font-format()` and `at-rule()` test supported features. A named condition is tested by placing its name inside parentheses:

```
@supports (color: navy) and selector(chapter > title) {
    chapter > title { color: navy; }
}

@supports-condition --has-navy {
    color: navy;
}

@supports (--has-navy) {
    chapter { color: navy; }
}
```

`@when` conditions combine `media()` and `supports()` tests with `and`, `or` and `not`. Parentheses group mixed operators:

```
@when media(width >= 6in) and supports(color: navy) {
    chapter { color: navy; }
}
```

Use `@else` immediately after an `@when`, `@media`, `@supports` or `@container` branch. An unrelated rule ends the chain.

Media features

Media feature values describe the page or PDF environment used for conditional rule evaluation. They do not describe a screen.

Name	Description	Values	Default	Specification
`any-hover`	Whether any pointing device can hover.	`none` | `hover`	`none`	Media Queries 5
`any-pointer`	The accuracy of any available pointing device.	`none` | `coarse` | `fine`	`none`	Media Queries 5
`aspect-ratio`	The ratio of page width to page height.	`<ratio>`	A4 page ratio	Media Queries 5
`color`	Bits per color component in the output.	`<integer>`	`8`	Media Queries 5
`color-gamut`	The approximate output color gamut.	`srgb` | `p3` | `rec2020`	`srgb`	Media Queries 5
`color-index`	Number of entries in the output color lookup table.	`<integer>`	`0`	Media Queries 5
`device-aspect-ratio`	The ratio of output width to output height.	`<ratio>`	A4 page ratio	Media Queries 5
`device-height`	Output height.	`<length>`	A4 page height	Media Queries 5
`device-width`	Output width.	`<length>`	A4 page width	Media Queries 5
`display-mode`	The presentation mode.	`fullscreen` | `standalone` | `minimal-ui` | `browser` | `picture-in-picture`	Not set	Media Queries 5
`dynamic-range`	The output's dynamic range.	`standard` | `high`	`standard`	Media Queries 5
`environment-blending`	Whether output blends with the surrounding environment.	`opaque` | `additive` | `subtractive`	`opaque`	Media Queries 5

Name	Description	Values	Default	Specification
`forced-colors`	Whether the output forces a limited color palette.	`none` | `active`	`none`	Media Queries 5
`grid`	Whether the output is grid-based.	`<mq-boolean>` (`0` or `1`)	`0`	Media Queries 5
`height`	Page height.	`<length>`	A4 page height	Media Queries 5
`horizontal-viewport-segments`	Number of horizontal viewport segments.	`<integer>`	`1`	Media Queries 5
`hover`	Whether the primary pointing device can hover.	`none` | `hover`	`none`	Media Queries 5
`inverted-colors`	Whether the output inverts colors.	`none` | `inverted`	`none`	Media Queries 5
`monochrome`	Number of bits per pixel in a monochrome output.	`<integer>`	`0`	Media Queries 5
`nav-controls`	Availability of navigation controls.	`none` | `back`	`none`	Media Queries 5
`orientation`	Whether the page is portrait or landscape.	`portrait` | `landscape`	`portrait`	Media Queries 5
`overflow-block`	How the output handles block-axis overflow.	`none` | `scroll` | `paged`	`paged`	Media Queries 5
`overflow-inline`	How the output handles inline-axis overflow.	`none` | `scroll`	`none`	Media Queries 5
`pointer`	The accuracy of the primary pointing device.	`none` | `coarse` | `fine`	`none`	Media Queries 5
`prefers-color-scheme`	The preferred color scheme.	`light` | `dark`	`light`	Media Queries 5

Name	Description	Values	Default	Specification
`prefers-contrast`	The preferred contrast level.	`no-preference` | `less` | `more` | `custom`	`no-preference`	Media Queries 5
`prefers-reduced-data`	Whether the user prefers reduced data use.	`no-preference` | `reduce`	`no-preference`	Media Queries 5
`prefers-reduced-motion`	Whether the user prefers reduced motion.	`no-preference` | `reduce`	`no-preference`	Media Queries 5
`prefers-reduced-transparency`	Whether the user prefers reduced transparency.	`no-preference` | `reduce`	`no-preference`	Media Queries 5
`resolution`	Output pixel density.	`<resolution>` | `infinite`	`infinite`	Media Queries 5
`scan`	The output's scanning process.	`interlace` | `progressive`	Not set	Media Queries 5
`scripting`	Whether scripting is available.	`none` | `initial-only` | `enabled`	`none`	Media Queries 5
`update`	How frequently the output can update.	`none` | `slow` | `fast`	`none`	Media Queries 5
`vertical-viewport-segments`	Number of vertical viewport segments.	`<integer>`	`1`	Media Queries 5
`video-color-gamut`	The color gamut of the output video.	`srgb` | `p3` | `rec2020`	`srgb`	Media Queries 5
`video-dynamic-range`	The dynamic range of the output video.	`standard` | `high`	`standard`	Media Queries 5
`width`	Page width.	`<length>`	A4 page width	Media Queries 5

- `<ratio>` is a ratio of two numbers.
- `<integer>` is a whole number.
- `<length>` is a CSS length, such as `12pt` or `2em`.
- `<mq-boolean>` is an integer with the value `0` or `1`.

- `<resolution>` is a pixel density, such as `300dpi`.

Container size features

A size query in `@container` tests the size of the nearest ancestor that has `container-type`. An ancestor with `size` answers all these features, and an ancestor with `inline-size` answers only the features of the inline axis.

Name	Description	Values	Default	Specification
`width`	The width of the content box of the container.	A length, alone or in a comparison or a range	The size of the container	CSS Conditional 5 §6.1
`height`	The height of the content box of the container.	A length, alone or in a comparison or a range	The size of the container	CSS Conditional 5 §6.1
`inline-size`	The size of the content box of the container along the lines of text.	A length, alone or in a comparison or a range	The size of the container	CSS Conditional 5 §6.1
`block-size`	The size of the content box of the container across the lines of text.	A length, alone or in a comparison or a range	The size of the container	CSS Conditional 5 §6.1
`aspect-ratio`	The ratio of the width to the height of the container.	A ratio, alone or in a comparison or a range	The ratio of the container	CSS Conditional 5 §6.1
`orientation`	Whether the container is wider than high.	`landscape` | `portrait`	The orientation of the container	CSS Conditional 5 §6.1

Container style queries

A container style query uses `style()` to test a computed style or custom property on an ancestor. See Conditional rules.

Margins, padding and borders

This reference lists the properties that CSSMedia uses for the margins, padding, borders and rounded corners of a box.

Margins

Name	Description	Values	Default	Specification
`margin`	Sets the space outside the four sides of the box. One value applies to all sides. Two values set top and bottom, then left and right. Three set top, then left and right, then bottom. Four set top, right, bottom and left.	One to four lengths or percentages, or `auto`	`0`	CSS Box Model 4 §3.2
`margin-top`	Sets the space above the box.	A length, a percentage or `auto`	`0`	CSS Box Model 4 §3.1
`margin-right`	Sets the space at the right of the box.	A length, a percentage or `auto`	`0`	CSS Box Model 4 §3.1
`margin-bottom`	Sets the space below the box.	A length, a percentage or `auto`	`0`	CSS Box Model 4 §3.1
`margin-left`	Sets the space at the left of the box.	A length, a percentage or `auto`	`0`	CSS Box Model 4 §3.1
`margin-block`	Sets the margins at the start and the end of the block axis.	One or two lengths, percentages or `auto`	`0`	CSS Logical Properties 1 §4.2
`margin-block-start`	Sets the margin at the start of the block axis.	A length, a percentage or `auto`	`0`	CSS Logical Properties 1 §4.2
`margin-block-end`	Sets the margin at the end of the block axis.	A length, a percentage or `auto`	`0`	CSS Logical Properties 1 §4.2
`margin-inline`	Sets the margins at the start and the end of the inline axis.	One or two lengths, percentages or `auto`	`0`	CSS Logical Properties 1 §4.2
`margin-inline-start`	Sets the margin at the start of the inline axis.	A length, a percentage or `auto`	`0`	CSS Logical Properties 1 §4.2
`margin-inline-end`	Sets the margin at the end of the inline axis.	A length, a percentage or `auto`	`0`	CSS Logical Properties 1 §4.2

Name	Description	Values	Default	Specification
`margin-break`	Chooses what happens to the margins of a box where a page or column break touches them.	`auto`, `keep` or `discard`	`auto`	CSS Fragmentation 4 §5.2
`margin-trim`	Removes the margins of the children that touch the edges of the container.	`none`, `block`, `inline`, or one or more of `block-start`, `block-end`, `inline-start` and `inline-end`	`none`	CSS Box Model 4 §3.3

Padding

Name	Description	Values	Default	Specification
`padding`	Sets the space between the content and the border on the four sides. The values are read like those of `margin`.	One to four non-negative lengths or percentages	`0`	CSS Box Model 4 §4.2
`padding-top`	Sets the space between the content and the top border.	A non-negative length or percentage	`0`	CSS Box Model 4 §4.1
`padding-right`	Sets the space between the content and the right border.	A non-negative length or percentage	`0`	CSS Box Model 4 §4.1
`padding-bottom`	Sets the space between the content and the bottom border.	A non-negative length or percentage	`0`	CSS Box Model 4 §4.1
`padding-left`	Sets the space between the content and the left border.	A non-negative length or percentage	`0`	CSS Box Model 4 §4.1
`padding-block`	Sets the padding at the start and the end of the block axis.	One or two non-negative lengths or percentages	`0`	CSS Logical Properties 1 §4.4
`padding-block-start`	Sets the padding at the start of the block axis.	A non-negative length or percentage	`0`	CSS Logical Properties 1 §4.4
`padding-block-end`	Sets the padding at the end of the block axis.	A non-negative length or percentage	`0`	CSS Logical Properties 1 §4.4

Name	Description	Values	Default	Specification
`padding-inline`	Sets the padding at the start and the end of the inline axis.	One or two non-negative lengths or percentages	`0`	CSS Logical Properties 1 §4.4
`padding-inline-start`	Sets the padding at the start of the inline axis.	A non-negative length or percentage	`0`	CSS Logical Properties 1 §4.4
`padding-inline-end`	Sets the padding at the end of the inline axis.	A non-negative length or percentage	`0`	CSS Logical Properties 1 §4.4

Borders

Name	Description	Values	Default	Specification
`border`	Sets the width, style and color of the four borders. It also resets `border-image`.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4
`border-width`	Sets the width of the four borders. The values are read like those of `margin`.	One to four widths	`medium`	CSS Borders 4 §2.3
`border-style`	Sets the line style of the four borders. The values are read like those of `margin`.	One to four styles	`none`	CSS Borders 4 §2.2
`border-color`	Sets the color of the four borders. The values are read like those of `margin`.	One to four colors	`currentcolor`	CSS Borders 4 §2.1
`border-top`	Sets the width, style and color of the top border.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4
`border-right`	Sets the width, style and color of the right border.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4
`border-bottom`	Sets the width, style and color of the bottom border.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4

Name	Description	Values	Default	Specification
`border-left`	Sets the width, style and color of the left border.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4
`border-block`	Sets the width, style and color of the borders at the start and the end of the block axis.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4
`border-block-start`	Sets the width, style and color of the border at the start of the block axis.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4
`border-block-end`	Sets the width, style and color of the border at the end of the block axis.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4
`border-inline`	Sets the width, style and color of the borders at the start and the end of the inline axis.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4
`border-inline-start`	Sets the width, style and color of the border at the start of the inline axis.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4
`border-inline-end`	Sets the width, style and color of the border at the end of the inline axis.	A width, a style and a color, in any order	See the individual properties	CSS Borders 4 §2.4
`border-top-width`	Sets the width of the top border.	`thin`, `medium`, `thick` or a non-negative length	`medium`	CSS Borders 4 §2.3
`border-right-width`	Sets the width of the right border.	`thin`, `medium`, `thick` or a non-negative length	`medium`	CSS Borders 4 §2.3
`border-bottom-width`	Sets the width of the bottom border.	`thin`, `medium`, `thick` or a non-negative length	`medium`	CSS Borders 4 §2.3
`border-left-width`	Sets the width of the left border.	`thin`, `medium`, `thick` or a non-negative length	`medium`	CSS Borders 4 §2.3

Name	Description	Values	Default	Specification
`border-block-width`	Sets the width of the borders at the start and the end of the block axis.	One or two widths	`medium`	CSS Borders 4 §2.3
`border-block-start-width`	Sets the width of the border at the start of the block axis.	`thin`, `medium`, `thick` or a non-negative length	`medium`	CSS Borders 4 §2.3
`border-block-end-width`	Sets the width of the border at the end of the block axis.	`thin`, `medium`, `thick` or a non-negative length	`medium`	CSS Borders 4 §2.3
`border-inline-width`	Sets the width of the borders at the start and the end of the inline axis.	One or two widths	`medium`	CSS Borders 4 §2.3
`border-inline-start-width`	Sets the width of the border at the start of the inline axis.	`thin`, `medium`, `thick` or a non-negative length	`medium`	CSS Borders 4 §2.3
`border-inline-end-width`	Sets the width of the border at the end of the inline axis.	`thin`, `medium`, `thick` or a non-negative length	`medium`	CSS Borders 4 §2.3
`border-top-style`	Sets the line style of the top border.	`none`, `hidden`, `dotted`, `dashed`, `solid`, `double`, `groove`, `ridge`, `inset` or `outset`	none	CSS Borders 4 §2.2
`border-right-style`	Sets the line style of the right border.	The values of `border-top-style`	none	CSS Borders 4 §2.2
`border-bottom-style`	Sets the line style of the bottom border.	The values of `border-top-style`	none	CSS Borders 4 §2.2
`border-left-style`	Sets the line style of the left border.	The values of `border-top-style`	none	CSS Borders 4 §2.2

Name	Description	Values	Default	Specification
`border-block-style`	Sets the line style of the borders at the start and the end of the block axis.	One or two styles	none	CSS Borders 4 §2.2
`border-block-start-style`	Sets the line style of the border at the start of the block axis.	The values of `border-top-style`	none	CSS Borders 4 §2.2
`border-block-end-style`	Sets the line style of the border at the end of the block axis.	The values of `border-top-style`	none	CSS Borders 4 §2.2
`border-inline-style`	Sets the line style of the borders at the start and the end of the inline axis.	One or two styles	none	CSS Borders 4 §2.2
`border-inline-start-style`	Sets the line style of the border at the start of the inline axis.	The values of `border-top-style`	none	CSS Borders 4 §2.2
`border-inline-end-style`	Sets the line style of the border at the end of the inline axis.	The values of `border-top-style`	none	CSS Borders 4 §2.2
`border-top-color`	Sets the color of the top border.	A color	`currentcolor`	CSS Borders 4 §2.1
`border-right-color`	Sets the color of the right border.	A color	`currentcolor`	CSS Borders 4 §2.1
`border-bottom-color`	Sets the color of the bottom border.	A color	`currentcolor`	CSS Borders 4 §2.1
`border-left-color`	Sets the color of the left border.	A color	`currentcolor`	CSS Borders 4 §2.1
`border-block-color`	Sets the color of the borders at the start and the end of the block axis.	One or two colors	`currentcolor`	CSS Borders 4 §2.1

Name	Description	Values	Default	Specification
`border-block-start-color`	Sets the color of the border at the start of the block axis.	A color	`currentcolor`	CSS Borders 4 §2.1
`border-block-end-color`	Sets the color of the border at the end of the block axis.	A color	`currentcolor`	CSS Borders 4 §2.1
`border-inline-color`	Sets the color of the borders at the start and the end of the inline axis.	One or two colors	`currentcolor`	CSS Borders 4 §2.1
`border-inline-start-color`	Sets the color of the border at the start of the inline axis.	A color	`currentcolor`	CSS Borders 4 §2.1
`border-inline-end-color`	Sets the color of the border at the end of the inline axis.	A color	`currentcolor`	CSS Borders 4 §2.1

Rounded corners

Name	Description	Values	Default	Specification
`border-radius`	Sets the radius of the four corners. Values after a slash set the vertical radii.	One to four non-negative lengths or percentages, optionally followed by `/` and one to four more	`0`	CSS Borders 4 §3.5
`border-top-left-radius`	Sets the radius of the top left corner. One value is a circle and two values are an ellipse.	One or two non-negative lengths or percentages	`0`	CSS Borders 4 §3.5
`border-top-right-radius`	Sets the radius of the top right corner.	One or two non-negative lengths or percentages	`0`	CSS Borders 4 §3.5
`border-bottom-right-radius`	Sets the radius of the bottom right corner.	One or two non-negative lengths or percentages	`0`	CSS Borders 4 §3.5

Name	Description	Values	Default	Specification
`border-bottom-left-radius`	Sets the radius of the bottom left corner.	One or two non-negative lengths or percentages	`0`	CSS Borders 4 §3.5
`border-start-start-radius`	Sets the radius of the corner at the start of the block axis and the start of the inline axis.	One or two non-negative lengths or percentages	`0`	CSS Borders 4 §3.5
`border-start-end-radius`	Sets the radius of the corner at the start of the block axis and the end of the inline axis.	One or two non-negative lengths or percentages	`0`	CSS Borders 4 §3.5
`border-end-start-radius`	Sets the radius of the corner at the end of the block axis and the start of the inline axis.	One or two non-negative lengths or percentages	`0`	CSS Borders 4 §3.5
`border-end-end-radius`	Sets the radius of the corner at the end of the block axis and the end of the inline axis.	One or two non-negative lengths or percentages	`0`	CSS Borders 4 §3.5
`border-top-radius`	Sets the radius of the two top corners.	One or two non-negative lengths or percentages, optionally with a slash	`0`	CSS Borders 4 §3.5
`border-right-radius`	Sets the radius of the two right corners.	One or two non-negative lengths or percentages, optionally with a slash	`0`	CSS Borders 4 §3.5
`border-bottom-radius`	Sets the radius of the two bottom corners.	One or two non-negative lengths or percentages, optionally with a slash	`0`	CSS Borders 4 §3.5
`border-left-radius`	Sets the radius of the two left corners.	One or two non-negative lengths or percentages, optionally with a slash	`0`	CSS Borders 4 §3.5
`border-block-start-radius`	Sets the radius of the two corners at the start of the block axis.	One or two non-negative lengths or percentages, optionally with a slash	`0`	CSS Borders 4 §3.5

Name	Description	Values	Default	Specification
`border-block-end-radius`	Sets the radius of the two corners at the end of the block axis.	One or two non-negative lengths or percentages, optionally with a slash	`0`	CSS Borders 4 §3.5
`border-inline-start-radius`	Sets the radius of the two corners at the start of the inline axis.	One or two non-negative lengths or percentages, optionally with a slash	`0`	CSS Borders 4 §3.5
`border-inline-end-radius`	Sets the radius of the two corners at the end of the inline axis.	One or two non-negative lengths or percentages, optionally with a slash	`0`	CSS Borders 4 §3.5
`corner`	Sets the radius and the shape of the four corners.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-top-left`	Sets the radius and the shape of the top left corner.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-top-right`	Sets the radius and the shape of the top right corner.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-bottom-right`	Sets the radius and the shape of the bottom right corner.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-bottom-left`	Sets the radius and the shape of the bottom left corner.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-start-start`	Sets the radius and the shape of the corner at the start of the block axis and the start of the inline axis.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-start-end`	Sets the radius and the shape of the corner at the start of the block axis and the end of the inline axis.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9

Name	Description	Values	Default	Specification
`corner-end-start`	Sets the radius and the shape of the corner at the end of the block axis and the start of the inline axis.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-end-end`	Sets the radius and the shape of the corner at the end of the block axis and the end of the inline axis.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-top`	Sets the radius and the shape of the two top corners.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-right`	Sets the radius and the shape of the two right corners.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-bottom`	Sets the radius and the shape of the two bottom corners.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-left`	Sets the radius and the shape of the two left corners.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-block-start`	Sets the radius and the shape of the two corners at the start of the block axis.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-block-end`	Sets the radius and the shape of the two corners at the end of the block axis.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-inline-start`	Sets the radius and the shape of the two corners at the start of the inline axis.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9
`corner-inline-end`	Sets the radius and the shape of the two corners at the end of the inline axis.	A radius and a shape	See the individual properties	CSS Borders 4 §3.9

Box sizes

This reference lists the properties that CSSMedia uses to size a box.

Preferred sizes

Name	Description	Values	Default	Specification
`box-sizing`	Chooses what `width` and `height` measure. `content-box` measures the content, and the padding and the border are added. `border-box` measures the box through the border.	`content-box` or `border-box`	`content-box`	CSS Sizing 3 §3.3
`width`	Sets the preferred width of the box.	`auto`, a non-negative length or percentage, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()` or `anchor-size()`	`auto`	CSS Sizing 3 §3.1
`height`	Sets the preferred height of the box.	The values of `width`	`auto`	CSS Sizing 3 §3.1
`inline-size`	Sets the preferred size along the inline axis, which is the width in horizontal text.	The values of `width`	`auto`	CSS Logical Properties 1 §4.1
`block-size`	Sets the preferred size along the block axis, which is the height in horizontal text.	The values of `width`	`auto`	CSS Logical Properties 1 §4.1
`size`	Sets `width` and `height`. One value applies to both.	One or two values of `width`	`auto`	CSS Sizing 4 §3.1

Minimum and maximum sizes

Name	Description	Values	Default	Specification
`min-width`	Sets the smallest width of the box.	The values of `width`	`auto`	CSS Sizing 3 §3.1
`min-height`	Sets the smallest height of the box.	The values of `width`	`auto`	CSS Sizing 3 §3.1

Name	Description	Values	Default	Specification
`min-inline-size`	Sets the smallest size along the inline axis.	The values of `width`	`auto`	CSS Logical Properties 1 §4.1
`min-block-size`	Sets the smallest size along the block axis.	The values of `width`	`auto`	CSS Logical Properties 1 §4.1
`min-size`	Sets `min-width` and `min-height`. One value applies to both.	One or two values of `min-width`	`auto`	CSS Sizing 4 §3.1
`max-width`	Sets the largest width of the box.	none, or the values of `width` except `auto`	none	CSS Sizing 3 §3.1
`max-height`	Sets the largest height of the box.	The values of `max-width`	none	CSS Sizing 3 §3.1
`max-inline-size`	Sets the largest size along the inline axis.	The values of `max-width`	none	CSS Logical Properties 1 §4.1
`max-block-size`	Sets the largest size along the block axis.	The values of `max-width`	none	CSS Logical Properties 1 §4.1
`max-size`	Sets `max-width` and `max-height`. One value applies to both.	One or two values of `max-width`	none	CSS Sizing 4 §3.1

Scaling

Name	Description	Values	Default	Specification
`zoom`	Scales all the lengths of an element, such as its font size, margins and width, by a factor. `2` and `200%` double them. The property is not inherited.	A number or a percentage of 0 or more	`1`	CSS Viewport 1 §4

Aspect ratio and intrinsic sizes

Name	Description	Values	Default	Specification
`aspect-ratio`	Gives the box a preferred ratio between its width and its height. The first number is the width and the second is the height. A single number is a ratio to 1.	`auto`, a non-negative number or two non-negative numbers separated by `/`, such as `16 / 9`. `auto` can come before or after the ratio.	`auto`	CSS Sizing 4 §4.1
`contain-intrinsic-width`	Sets the width that a box with size containment has instead of the size of its content.	An optional `auto`, then none or a non-negative length	none	CSS Sizing 4 §5.2
`contain-intrinsic-height`	Sets the height that a box with size containment has instead of the size of its content.	The values of `contain-intrinsic-width`	none	CSS Sizing 4 §5.2
`contain-intrinsic-inline-size`	Sets the inline size that a box with size containment has instead of the size of its content.	The values of `contain-intrinsic-width`	none	CSS Sizing 4 §5.2
`contain-intrinsic-block-size`	Sets the block size that a box with size containment has instead of the size of its content.	The values of `contain-intrinsic-width`	none	CSS Sizing 4 §5.2
`contain-intrinsic-size`	Sets `contain-intrinsic-width` and `contain-intrinsic-height`. One value applies to both.	One or two values of `contain-intrinsic-width`	none	CSS Sizing 4 §5.2
`frame-sizing`	Chooses the size of a replaced element that is used for its sizing.	`auto`, `content-width`, `content-height`, `content-block-size` or `content-inline-size`	`auto`	CSS Sizing 4 §5.3

Name	Description	Values	Default	Specification
`min-intrinsic-sizing`	Chooses when the minimum size that a box contributes to its container is compressed.	`legacy`, `zero-if-scroll`, `zero-if-extrinsic` or both of the last two	`legacy`	CSS Sizing 4 §5.5

Backgrounds

This reference lists the properties that CSSMedia uses to paint the background of a box.

Color, images and shorthands

Name	Description	Values	Default	Specification
`background`	Sets all the background properties in one declaration. Layers are separated by commas, and the color goes in the last layer. It resets the properties that you leave out.	One or more layers, each with an image, a position, a size after `/`, a repeat style, an attachment and one or two boxes; and a color in the last layer	See the individual properties	CSS Backgrounds 4 §2.12
`background-tbd`	Sets the background layers like `background`, but takes no color and does not reset it.	One or more layers	See the individual properties	CSS Backgrounds 4 §2.11
`background-color`	Sets the solid color painted below the images.	A color	`transparent`	CSS Backgrounds 4 §2.1
`background-image`	Sets the images of the layers. The first image is the top layer.	A comma-separated list of none or images: `url()`, gradients, `image()`, `image-set()` and `cross-fade()`	none	CSS Backgrounds 4 §2.2

Position, size and repetition

Name	Description	Values	Default	Specification
`background-position`	Sets where each image is placed in its positioning area.	One to four values: `left`, `center`, `right`, `top`, `bottom`, lengths and percentages. A side keyword can be followed by an offset.	`0% 0%`	CSS Backgrounds 4 §2.7
`background-position-x`	Sets the horizontal position of each image.	`center`, `left`, `right`, `x-start`, `x-end`, a length or percentage, or a side keyword with an offset	`0%`	CSS Backgrounds 4 §2.6
`background-position-y`	Sets the vertical position of each image.	`center`, `top`, `bottom`, `y-start`, `y-end`, a length or percentage, or a side keyword with an offset	`0%`	CSS Backgrounds 4 §2.6
`background-position-inline`	Sets the position of each image along the inline axis.	`center`, `start` or end, a length or percentage, or a keyword with an offset	`0%`	CSS Backgrounds 4 §2.6
`background-position-block`	Sets the position of each image along the block axis.	The values of `background-position-inline`	`0%`	CSS Backgrounds 4 §2.6
`background-size`	Sets the size of each image. `cover` makes the image cover the area, and `contain` makes the whole image fit in it.	`cover`, `contain`, or one or two values that are `auto` or non-negative lengths or percentages	`auto`	CSS Backgrounds 4 §2.10
`background-repeat`	Sets how each image repeats on the two axes. One value applies to both axes, and two values are the horizontal and the vertical ones.	`repeat-x`, `repeat-y`, `repeat-inline`, `repeat-block`, or one or two of `repeat`, `no-repeat`, `space` and `round`	`repeat`	CSS Backgrounds 4 §2.4

Name	Description	Values	Default	Specification
`background-repeat-x`	Sets how each image repeats on the horizontal axis.	`repeat`, `no-repeat`, `space` or `round`	`repeat`	CSS Backgrounds 4 §2.3
`background-repeat-y`	Sets how each image repeats on the vertical axis.	The values of `background-repeat-x`	`repeat`	CSS Backgrounds 4 §2.3
`background-repeat-inline`	Sets how each image repeats along the inline axis.	The values of `background-repeat-x`	`repeat`	CSS Backgrounds 4 §2.3
`background-repeat-block`	Sets how each image repeats along the block axis.	The values of `background-repeat-x`	`repeat`	CSS Backgrounds 4 §2.3
`background-origin`	Chooses the box that is the positioning area of the images.	`border-box`, `padding-box` or `content-box`	`padding-box`	CSS Backgrounds 4 §2.9
`background-clip`	Chooses the area that the background is painted in.	`border-box`, `padding-box`, `content-box`, `border-area`, `text`, or `border-area` and `text` together	`border-box`	CSS Backgrounds 4 §2.8
`background-attachment`	Chooses what the images are attached to. In a PDF, `fixed` attaches them to the page, so that they are drawn again on every page.	`scroll`, `local` or `fixed`	`scroll`	CSS Backgrounds 4 §2.5

Blending

Name	Description	Values	Default	Specification
`background-blend-mode`	Sets how each image layer blends with the layers below it and with the background color.	`normal`, `multiply`, `screen`, `overlay`, `darken`, `lighten`, `color-dodge`, `color-burn`, `hard-light`, `soft-light`, `difference`, `exclusion`, hue, `saturation`, `color` or `luminosity`; a comma-separated list	`normal`	Compositing 1 §3.4.3

Shadows, opacity, transforms and masks

This reference lists the properties that make an element transparent, move, turn or scale it, clip it or mask it, and mix it with what is behind it.

Opacity, clipping, masking and transform properties

Name	Description	Values	Default	Specification
`opacity`	Makes an element and its children transparent as a group.	A number from 0 to 1, or a percentage	`1`	CSS Color 4 §3.3
`clip-path`	Shows only the part of an element inside an area.	none, `url(#id)` of a `clipPath`, or a basic shape, with `fill-box`, `stroke-box` or `view-box`	none	CSS Masking 1 §5.1
`clip-rule`	Chooses which areas of a shape inside a `clipPath` are inside it.	`nonzero` or evenodd	`nonzero`	CSS Masking 1 §6.2
`mask`	Makes parts of an element transparent with a `mask` element.	none or `url(#id)` of a `mask` element	none	CSS Masking 1 §7.9
`mask-type`	Chooses whether a `mask` element uses the luminance or the opacity of what it holds.	`luminance` or `alpha`	`luminance`	CSS Masking 1 §9.2

Name	Description	Values	Default	Specification
`transform`	Moves, turns, scales or skews an element and its children.	none or a list of transform functions, 2D or 3D	none	CSS Transforms 1 §4
`transform-origin`	Sets the point that `transform` turns and scales around. In a drawing it is the origin of the drawing.	One to three lengths, percentages or keywords	`0 0`	CSS Transforms 1 §5
`translate`, `rotate`, `scale`	Move, turn or scale an element, each in a property of its own.	none or the arguments of the function of the same name	none	CSS Transforms 2 §5

Blending

Name	Description	Values	Default	Specification
`mix-blend-mode`	Sets how an element blends with what is behind it.	The values of `background-blend-mode`	`normal`	Compositing 1 §3.4.1
`isolation`	Chooses whether an element makes a group of its own for blending.	`auto` or `isolate`	`auto`	Compositing 1 §3.4.2

Display and positioning

The `display` property determines the kind of box an element generates. Positioning properties describe how a box is placed relative to its normal position or containing block.

Name	Description	Values	Default	Specification
`display`	Selects whether an element generates a box and describes its outer and inner display types.	none, `contents`, or combinations of outer types (`block`, `inline`, `run-in`), inner types (`flow`, `flow-root`, `table`, `flex`, `grid`, `ruby`), and `list-item`; also internal table and ruby types, legacy combinations such as `inline-block`, `inline-flex`, and `inline-grid`, plus		CSS Display 4 §2

Name	Description	Values	Default	Specification
		`math`, `grid-lanes`, and `inline-grid-lanes`		
`visibility`	Controls whether a box is visible while preserving its place in layout.	`visible`, `hidden`, `force-hidden`, or `collapse`		CSS Display 4 §5
`order`	Changes the order of flex and grid items for layout and painting.	An integer		CSS Display 4 §3
`reading-flow`	Chooses the reading order of children in a block, flex, or grid container.	`normal`, `source-order`, `flex-visual`, `flex-flow`, `grid-rows`, `grid-columns`, or `grid-order`		CSS Display 4 §4
`reading-order`	Assigns an integer ordering value to a direct child of a reading-flow container.	An integer		CSS Display 4 §4.1
`position`	Chooses the positioning scheme for an element.	`static`, `relative`, `absolute`, `sticky`, `fixed`, or `running()`		CSS Position 3 §2; CSS GCPM 3 §1.2.1
`top`, `right`, `bottom`, `left`	Set an inset from the corresponding side of the containing block for a positioned element.	`auto`, a length or percentage, `anchor()`, or `anchor-size()`		CSS Position 3 §3.1
`inset`	Sets the four physical insets.	One to four values accepted by `top`, `right`, `bottom`, and `left`		CSS Position 3 §3.2
`inset-block`, `inset-inline`	Set the start and end insets along the block or inline axis.	One or two values accepted by the corresponding logical inset properties		CSS Position 3 §3.2

Name	Description	Values	Default	Specification
`inset-block-start`, `inset-block-end`, `inset-inline-start`, `inset-inline-end`	Set one inset on the block or inline axis.	`auto`, a length or percentage, `anchor()`, or `anchor-size()`		CSS Position 3 §3.1
`z-index`	Sets a positioned box's stack level.	`auto` or an integer		CSS 2 §9.9
`float`	Takes a box out of the normal flow and places it at the side, at the top or at the bottom of its container. The value `footnote` makes the element a footnote.	`none`, `left`, `right`, `top`, `bottom`, `block-start`, `block-end`, `inline-start`, `inline-end`, `snap-block`, `snap-inline` or `footnote`	none	CSS Page Floats 3 §3.2

- A length is a size with a unit, such as `12pt` or `1em`. A percentage is relative to the containing block, as defined for the relevant inset.
- `anchor()` uses an anchor's side or center as an inset coordinate. `anchor-size()` uses a selected anchor's size. See CSS Anchor Positioning 1 and the `anchor-size()` function.
- `running()` names an element to move into a page margin box. See CSS GCPM 3.
- An integer is a whole number without a fractional part. Negative values, zero, and positive values are allowed for `order` and `z-index`; CSS Display defines how ordering is applied.

Tables

This reference lists the display values that make a table and the properties that style it.

Display values of tables

`display`	The element is	Specification
`table`	A table, as a block.	CSS Tables 3 §2
`inline-table`	A table that sits in a line of text.	CSS Tables 3 §2
`table-caption`	The caption of the table, above it or below it.	CSS Tables 3 §2
`table-header-group`	Rows that are always at the start of the table, and that can repeat on every page.	CSS Tables 3 §2

display	The element is	Specification
table-row-group	A group of rows.	CSS Tables 3 §2
table-footer-group	Rows that are always at the end of the table, and that can repeat on every page.	CSS Tables 3 §2
table-row	A row of cells.	CSS Tables 3 §2
table-cell	A cell, with the content of the table.	CSS Tables 3 §2
table-column	A column. It has no content, and it gives a width, a border or a background to the cells of a column.	CSS Tables 3 §2
table-column-group	A group of columns.	CSS Tables 3 §2

Table properties

Name	Description	Values	Default	Specification
table-layout	Chooses how the widths of the columns are found: from the content of all the cells, or from the first row and the column elements.	auto or fixed; fixed needs a width on the table	auto	CSS Tables 3 §3.5.1
border-collapse	Chooses between a border for each cell and one border shared by neighbors.	separate or collapse	separate	CSS Tables 3 §3.5.2
border-spacing	Sets the gap between the borders of neighbor cells when the borders are separate.	One length for both directions, or two lengths for the horizontal and the vertical gap	0	CSS Tables 3 §3.5.2.1
caption-side	Puts the caption above or below the table.	top or bottom	top	CSS Tables 3 §3.5.3
empty-cells	Chooses whether the borders and the background of a cell without content are shown, when the borders are separate.	show or hide	show	CSS Tables 3 §5.2

Name	Description	Values	Default	Specification
`vertical-align`	On a cell, sets where its content goes when the row is taller than the content.	`baseline`, `top`, `middle` or `bottom`	`baseline`	CSS Inline 3

Text and fonts

This reference lists the properties, at-rules and descriptors that CSSMedia uses to select, load and shape fonts.

Font selection and sizing

Name	Description	Values	Default	Specification
`font`	Sets the font style, variant, weight, width, size, line height and family in one declaration.	A system font keyword (`caption`, `icon`, `menu`, `message-box`, `small-caption` or `status-bar`), or optional style, `normal` or `small-caps`, weight and width, then the size, an optional `/` and line height, and the family	See the individual properties	CSS Fonts 4 §2.7
`font-family`	Lists the font families to try, in order. A family name with spaces needs quotes.	Family names, and `serif`, `sans-serif`, `monospace`, `system-ui`, `cursive`, `fantasy`, `math`, `ui-serif`, `ui-sans-serif`, `ui-monospace`, `ui-rounded` or `generic()` keywords	`sans-serif`, which is Helvetica	CSS Fonts 4 §2.1
`font-size`	Sets the size of the text.	A keyword from `xx-small` to `xxx-large`, `larger`, `smaller`, `math`, or a non-negative length or percentage	`medium`	CSS Fonts 4 §2.5

Name	Description	Values	Default	Specification
`font-size-adjust`	Scales the text so that a chosen metric of the font, such as the height of lowercase letters, has the same relative size in every font.	none, or an optional metric (`ex-height`, `cap-height`, `ch-width`, `ic-width` or `ic-height`) followed by `from-font` or a non-negative number	none	CSS Fonts 5 §3.4
`font-weight`	Selects the weight of the face.	`normal`, `bold`, `bolder`, `lighter` or a number from 1 to 1000	`normal`	CSS Fonts 4 §2.2
`font-width`	Selects the width of the face.	`normal`, a non-negative percentage, or `ultra-condensed`, `extra-condensed`, `condensed`, `semi-condensed`, `semi-expanded`, `expanded`, `extra-expanded` or `ultra-expanded`	`normal`	CSS Fonts 4 §2.3
`font-stretch`	Alias of `font-width`.	The values of `font-width`	`normal`	CSS Fonts 4 §2.3.1
`font-style`	Selects an upright, italic or oblique face.	`normal`, `italic`, `left`, `right`, or `oblique` with an optional angle from -90deg to 90deg	`normal`	CSS Fonts 4 §2.4
`font-synthesis`	Allows or forbids the faces that CSSMedia creates when the family has none: bold, slanted, small capitals, subscripts and superscripts.	none, or any of `weight`, `style`, `small-caps` and `position`	`weight style small-caps position`	CSS Fonts 4 §2.8.5
`font-synthesis-weight`	Allows or forbids synthesized bold.	`auto` or none	`auto`	CSS Fonts 4 §2.8.1
`font-synthesis-style`	Allows or forbids synthesized italic and oblique faces.	`auto`, none or `oblique-only`	`auto`	CSS Fonts 4 §2.8.2

Name	Description	Values	Default	Specification
`font-synthesis-small-caps`	Allows or forbids synthesized small capitals.	`auto` or `none`	`auto`	CSS Fonts 4 §2.8.3
`font-synthesis-position`	Allows or forbids synthesized subscripts and superscripts.	`auto` or `none`	`auto`	CSS Fonts 4 §2.8.4
`line-height`	Sets the height of a line of text.	`normal`, a non-negative number, or a non-negative length or percentage	`normal`	CSS Inline Layout 3 §5.1

Text color, alignment and lines

Name	Description	Values	Default	Specification
`color`	Sets the color of the text. Borders and other properties that use `currentcolor` take it too, and the property is inherited.	A color: a keyword, `#rgb` or `#rrggbb`, `rgb()`, `hsl()` or another color function, `currentcolor` or `transparent`	`CanvasText`, black	CSS Color 4 §3.2
`text-align`	Aligns the lines of a block. `justify` stretches every line but the last, and `justify-all` stretches the last one too. `start` and end follow the direction of the text.	`start`, `end`, `left`, `right`, `center`, `justify`, `justify-all` or `match-parent`	`start`	CSS Text 4 §7.1
`text-transform`	Changes the case of the text when it is drawn, without changing your XML. `full-width` replaces the letters and digits by their full-width forms, which CJK text uses.	none, or `capitalize`, `uppercase` or `lowercase`, with `full-width`	none	CSS Text 4 §2.1

Name	Description	Values	Default	Specification
`text-decoration`	Draws lines on the text. It is a shorthand for `text-decoration-line`, `text-decoration-style`, `text-decoration-color` and `text-decoration-thickness`. The lines run across the text of the children too.	A line: none or one or more of `underline`, `overline` and `line-through`; a style: `solid`, `double`, `dotted`, `dashed` or `wavy`; a color; a thickness	`none`	CSS Text Decoration 4 §2.6
`white-space`	Chooses whether spaces and line feeds collapse and whether lines wrap. It is a shorthand for `white-space-collapse` and `text-wrap-mode`. `pre` keeps spaces and line feeds and does not wrap, `pre-wrap` keeps them and wraps, `pre-line` keeps line feeds only, and `nowrap` collapses spaces and does not wrap. `break-spaces` keeps spaces and lets a line break after each.	`normal`, `pre`, `nowrap`, `pre-wrap`, `pre-line` or `break-spaces`	`normal`	CSS Text 4 §3
`overflow-wrap`	Lets a word that is longer than its line break at any letter, so that it does not overflow the box.	`normal`, `break-word` or `anywhere`	`normal`	CSS Text 4 §6.4

Font variants and features

Name	Description	Values	Default	Specification
`font-variant`	Shorthand for the `font-variant-*` properties.	`normal`, none, or values of the longhand properties	`normal`	CSS Fonts 4 §6.11

Name	Description	Values	Default	Specification
`font-variant-alternates`	Selects alternate glyphs that the font names, such as stylistic sets and swashes.	`normal`, or any of `stylistic()`, `historical-forms`, `styleset()`, `character-variant()`, `swash()`, `ornaments()` and `annotation()`	`normal`	CSS Fonts 4 §6.8
`font-variant-caps`	Selects small capitals and other capital forms.	`normal`, `small-caps`, `all-small-caps`, `petite-caps`, `all-petite-caps`, `unicase` or `titling-caps`	`normal`	CSS Fonts 4 §6.6
`font-variant-east-asian`	Selects glyph forms and widths for East Asian text.	`normal`, or any of `jis78`, `jis83`, `jis90`, `jis04`, `simplified`, `traditional`, `full-width`, `proportional-width` and `ruby`	`normal`	CSS Fonts 4 §6.10
`font-variant-emoji`	Chooses text or emoji presentation for characters that have both.	`normal`, `text`, `emoji` or `unicode`	`normal`	CSS Fonts 4 §9.3
`font-variant-ligatures`	Turns common, discretionary, historical and contextual ligatures on or off.	`normal`, `none`, or any of `common-ligatures`, `no-common-ligatures`, `discretionary-ligatures`, `no-discretionary-ligatures`, `historical-ligatures`, `no-historical-ligatures`, `contextual` and `no-contextual`	`normal`	CSS Fonts 4 §6.4
`font-variant-numeric`	Selects the shape and spacing of numbers, fractions and zeros.	`normal`, or any of `lining-nums`, `oldstyle-nums`, `proportional-nums`, `tabular-nums`, `diagonal-fractions`, `stacked-fractions`, `ordinal` and `slashed-zero`	`normal`	CSS Fonts 4 §6.7

Name	Description	Values	Default	Specification
`font-variant-position`	Selects subscript or superscript glyphs.	`normal`, `sub` or `super`	`normal`	CSS Fonts 4 §6.5
`font-kerning`	Turns the kerning of the font on or off.	`auto`, `normal` or `none`	`auto`	CSS Fonts 4 §6.3
`font-feature-settings`	Turns OpenType features on or off by their four-letter tags.	`normal`, or a comma-separated list of tags, each followed by an optional integer, on or `off`	`normal`	CSS Fonts 4 §6.12
`font-language-override`	Sets the OpenType language system of the text, in place of its language.	`normal` or a string	`normal`	CSS Fonts 4 §6.13

Variable and color fonts

Name	Description	Values	Default	Specification
`font-optical-sizing`	Sets the optical size axis of a variable font from the font size.	`auto` or `none`	`auto`	CSS Fonts 4 §8.1
`font-variation-settings`	Sets the axes of a variable font by their four-letter tags.	`normal`, or a comma-separated list of axis tags in quotes, each followed by a number	`normal`	CSS Fonts 4 §8.2
`font-palette`	Chooses the color palette of a color font.	`normal`, `light`, `dark`, the name of an `@font-palette-values` rule, or `palette-mix()`	`normal`	CSS Fonts 4 §9.1

The @font-face rule

Name	Description	Values	Default	Specification
`@font-face`	Makes a font face available to the stylesheet under a family name.	A block of the descriptors below		CSS Fonts 4 §4.1

Name	Description	Values	Default	Specification
`font-family` descriptor	Names the family of the face. This name hides an installed family with the same name.	A family name	Required	CSS Fonts 4 §4.2
`src` descriptor	Lists the sources of the face. CSSMedia uses the first one that it can load.	A comma-separated list of `url()` and `local()` sources, each with optional `format()` and `tech()` hints	Required	CSS Fonts 4 §4.3
`font-weight` descriptor	Gives the weight, or the range of weights, of the face.	`auto`, or one or two values, each `normal`, `bold` or a number from 1 to 1000	`auto`	CSS Fonts 4 §4.4
`font-width` descriptor	Gives the width, or the range of widths, of the face.	`auto`, or one or two values, each a width keyword or a non-negative percentage	`auto`	CSS Fonts 4 §4.4
`font-stretch` descriptor	Alias of the `font-width` descriptor.	The values of the `font-width` descriptor	`auto`	CSS Fonts 4 §4.4.1
`font-style` descriptor	Gives the style, or the range of oblique angles, of the face.	`auto`, `normal`, `italic`, `left`, `right`, or `oblique` with one or two angles from -90deg to 90deg	`auto`	CSS Fonts 4 §4.4
`font-size` descriptor	Limits the font sizes for which the face is used.	`auto` or one or two numbers	`auto`	CSS Fonts 5 §4.3
`font-named-instance` descriptor	Selects a named instance of a variable font.	`auto` or a string	`auto`	CSS Fonts 4 §4.7

Name	Description	Values	Default	Specification
`unicode-range` descriptor	Limits the characters for which the face is used.	A comma-separated list of code points, ranges and ranges with ? wildcards	`U+0-10FFFF`	CSS Fonts 4 §4.5
`font-feature-settings` descriptor	Sets the OpenType features that the face uses by default.	The values of the `font-feature-settings` property	`normal`	CSS Fonts 4 §4.6
`font-variation-settings` descriptor	Sets the variable font axes that the face uses by default.	The values of the `font-variation-settings` property	`normal`	CSS Fonts 4 §4.6
`font-language-override` descriptor	Sets the OpenType language system that the face uses by default.	`normal` or a string	`normal`	CSS Fonts 4 §4.10
`size-adjust` descriptor	Scales the glyphs and the metrics of the face.	A non-negative percentage	`100%`	CSS Fonts 5 §4.4
`ascent-override` descriptor	Replaces the ascent of the face.	One or two values, each `normal` or a non-negative percentage	`normal`	CSS Fonts 5 §4.5
`descent-override` descriptor	Replaces the descent of the face.	One or two values, each `normal` or a non-negative percentage	`normal`	CSS Fonts 5 §4.5
`line-gap-override` descriptor	Replaces the line gap of the face.	One or two values, each `normal` or a non-negative percentage	`normal`	CSS Fonts 5 §4.5
`superscript-position-override` descriptor	Replaces the position of the superscript glyphs.	One or two values, each `normal`, `from-font` or a percentage	`normal`	CSS Fonts 5 §4.6

Name	Description	Values	Default	Specification
`superscript-size-override` descriptor	Replaces the size of the superscript glyphs.	One or two values, each `normal`, `from-font` or a non-negative percentage	`normal`	CSS Fonts 5 §4.6
`subscript-position-override` descriptor	Replaces the position of the subscript glyphs.	One or two values, each `normal`, `from-font` or a percentage	`normal`	CSS Fonts 5 §4.6
`subscript-size-override` descriptor	Replaces the size of the subscript glyphs.	One or two values, each `normal`, `from-font` or a non-negative percentage	`normal`	CSS Fonts 5 §4.6

The @font-feature-values rule

Name	Description	Values	Default	Specification
`@font-feature-values`	Gives names to the alternate glyphs of one or more font families, for use in `font-variant-alternates`.	One or more family names, and a block with the rules below		CSS Fonts 4 §6.9
`@stylistic`	Names stylistic alternates (`salt`).	Names, each with one non-negative integer		CSS Fonts 4 §6.9.1
`@styleset`	Names stylistic sets (`ss01` to `ss99`).	Names, each with one or more integers from 1 to 99. 0 and larger values select no feature.		CSS Fonts 4 §6.9.2
`@character-variant`	Names character variants (`cv01` to `cv99`).	Names, each with one or two integers. The first, from 1 to 99, selects the feature, and the second sets its value. 0 and larger values select no feature.		CSS Fonts 4 §6.9.2
`@swash`	Names swashes (`swsh`, `cswh`).	Names, each with one non-negative integer		CSS Fonts 4 §6.9.1

Name	Description	Values	Default	Specification
`@ornaments`	Names ornaments (`ornm`).	Names, each with one non-negative integer		CSS Fonts 4 §6.9.1
`@annotation`	Names annotation forms (`nalt`).	Names, each with one non-negative integer		CSS Fonts 4 §6.9.1
`@historical-forms`	Names historical forms (`hist`).	Names, each with one or more non-negative integers		CSS Fonts 4 §6.9.1

The @font-palette-values rule

Name	Description	Values	Default	Specification
`@font-palette-values`	Defines a named color palette for a color font. Use its name in `font-palette`.	A name that starts with `--`, and a block of the descriptors below		CSS Fonts 4 §9.2
`font-family` descriptor	Names the families that the palette is for.	One or more family names	Required	CSS Fonts 4 §9.2.1
`base-palette` descriptor	Chooses the palette of the font that the new palette starts from.	`light`, `dark` or a non-negative integer	The first palette of the font	CSS Fonts 4 §9.2.2
`override-colors` descriptor	Replaces colors of the base palette.	A comma-separated list of pairs of a palette index and a color	No replacements	CSS Fonts 4 §9.2.3

Languages and hyphenation

This reference lists the property, the selector and the properties that CSSMedia uses for the language of the text, hyphenation and quotation marks.

Properties

Name	Description	Values	Default	Specification
`-cssmedia-lang`	Sets the language of the elements from a value of your XML, such as an attribute. CSSMedia extension.	none or a string with a language tag. `attr()` reads the tag from an attribute.	none	CSSMedia extension

Name	Description	Values	Default	Specification
`hyphens`	Chooses how words are broken at the end of a line. `manual` breaks only at the soft hyphens that the text has, `auto` also uses the hyphenation patterns of the language, and none never breaks inside a word.	none, `manual` or `auto`	`manual`	CSS Text 4 §6.3.1
`hyphenate-character`	Sets the string that CSSMedia writes at the end of a line that it breaks inside a word.	`auto` or a string. `auto` is U+2010 when the font has it, and U+002D otherwise.	`auto`	CSS Text 4 §6.3.2
`hyphenate-limit-chars`	Sets the shortest word that CSSMedia hyphenates, and the least number of letters before and after the hyphen.	One to three values, each `auto` or an integer. The values are the word length, the letters before and the letters after. A missing third value is the second, and a missing second value is `auto`.	`auto`	CSS Text 4 §6.3.4
`hyphenate-limit-lines`	Sets the largest number of consecutive lines of an element that end with a hyphen.	`no-limit` or an integer	`no-limit`	CSS Text 4 §6.3.5
`hyphenate-limit-last`	Stops CSSMedia from ending a line with a hyphen at the end of an element or before a page break. `always` applies to the last full line of the element and to the last line before a page break. `column` and page apply to the last line before a page break, and `spread` to the last line before a spread break.	none, `always`, `column`, page or `spread`	none	CSS Text 4 §6.3.5

Name	Description	Values	Default	Specification
`hyphenate-limit-zone`	Sets how much empty space a line can have before CSSMedia hyphenates a word to fill it. A percentage is relative to the width of the line.	A length or a percentage	`0`	CSS Text 4 §6.3.3
`quotes`	Selects the quotation marks used by generated `open-quote` and `close-quote` content. With `auto`, the marks depend on the language of the text.	`auto`, `none`, `match-parent`, or one or more pairs of quoted strings	`auto`	CSS Content 3 §2.4.1

Selector

Name	Description	Values	In a PDF	Specification
`:lang()`	Elements in a language. The language comes from `xml:lang` or the `-cssmedia-lang` property.	One or more language ranges, as identifiers or strings.	Matches	Selectors 4 §7.2

Page reference

Name	Description	Values	Default	Specification
`page`	Assigns a named page type to a box.	`auto` or a custom identifier	`auto`	CSS Paged Media 3, `page`
`@page`	Defines page-box declarations and may contain page-margin rules.	Optional comma-separated page selectors		CSS Paged Media 3, `@page`
`size`	Sets page-box size and orientation. `auto` uses A4.	`auto`; one or two lengths; or A3, A4, A5, B4, B5, `JIS-B4`, `JIS-B5`, `ledger`, `legal`, or `letter`, optionally combined with `portrait` or `landscape`	A4	CSS Paged Media 3, `size`

Name	Description	Values	Default	Specification
`bleed`	Sets the bleed extension beyond the page box. `auto` uses `6pt` with crop marks and `0` otherwise.	`auto` or a length	`0` (`auto`)	CSS Paged Media 3, `bleed`
`marks`	Requests crop marks, registration marks, or both outside the page box.	`none`, `crop`, `cross`, or both `crop` and `cross`	`none`	CSS Paged Media 3, `marks`
`page-orientation`	Rotates the laid-out page in the output.	`upright`, `rotate-left`, or `rotate-right`	`upright`	CSS Paged Media 3, `page-orientation`
`@top-left-corner`	Defines the fixed-size box at the intersection of the top and left page margins.			CSS Paged Media 3, `@top-left-corner`
`@top-left`	Defines the box between the top-left corner and top-center boxes.			CSS Paged Media 3, `@top-left`
`@top-center`	Defines the centered box in the top page margin.			CSS Paged Media 3, `@top-center`
`@top-right`	Defines the box between the top-center and top-right corner boxes.			CSS Paged Media 3, `@top-right`
`@top-right-corner`	Defines the fixed-size box at the intersection of the top and right page margins.			CSS Paged Media 3, `@top-right-corner`
`@right-top`	Defines the box between the top-right corner and right-middle boxes.			CSS Paged Media 3, `@right-top`

Name	Description	Values	Default	Specification
`@right-middle`	Defines the centered box in the right page margin.			CSS Paged Media 3, `@right-middle`
`@right-bottom`	Defines the box between the right-middle and bottom-right corner boxes.			CSS Paged Media 3, `@right-bottom`
`@bottom-right-corner`	Defines the fixed-size box at the intersection of the bottom and right page margins.			CSS Paged Media 3, `@bottom-right-corner`
`@bottom-right`	Defines the box between the bottom-center and bottom-right corner boxes.			CSS Paged Media 3, `@bottom-right`
`@bottom-center`	Defines the centered box in the bottom page margin.			CSS Paged Media 3, `@bottom-center`
`@bottom-left`	Defines the box between the bottom-left corner and bottom-center boxes.			CSS Paged Media 3, `@bottom-left`
`@bottom-left-corner`	Defines the fixed-size box at the intersection of the bottom and left page margins.			CSS Paged Media 3, `@bottom-left-corner`
`@left-bottom`	Defines the box between the left-middle and bottom-left corner boxes.			CSS Paged Media 3, `@left-bottom`
`@left-middle`	Defines the centered box in the left page margin.			CSS Paged Media 3, `@left-middle`

Name	Description	Values	Default	Specification
`@left-top`	Defines the box between the top-left corner and left-middle boxes.			CSS Paged Media 3, `@left-top`
`@footnote`	Styles the footnote area at the bottom of the page. It goes inside an `@page` rule.	A block of declarations, such as borders, margins, padding and `max-height`.		CSS GCPM 3 §2.4

Breaks

Name	Description	Values	Default	Specification
`break-before`	Forces or avoids a break before a block, a row or a row group. `page`, `always` and `all` start a new page. `left` and `verso` start the next left page, and `right` and `recto` the next right page, with a blank page when needed. `avoid` and `avoid-page` move the break to an earlier place. `column` breaks only in columns.	`auto`, `avoid`, `avoid-page`, `page`, `always`, `all`, `left`, `right`, `recto`, `verso`, `column` or `avoid-column`	`auto`	CSS Fragmentation 4 §3.1
`break-after`	Forces or avoids a break after a block, a row or a row group, with the same values as `break-before`.	As `break-before`	`auto`	CSS Fragmentation 4 §3.1
`break-inside`	Avoids a break inside a box, so that it stays on one page when it fits on one. A table header or footer group with `avoid` repeats on each page of the table.	`auto`, `avoid`, `avoid-page` or `avoid-column`	`auto`	CSS Fragmentation 4 §3.2
`orphans`	Sets how many lines of a paragraph must stay at the bottom of a page. If fewer fit, the whole paragraph goes to the next page.	An integer of 1 or more	2	CSS Fragmentation 4 §3.3
`widows`	Sets how many lines of a paragraph must go to the top of the next page. If fewer would, lines from the previous page go with them.	An integer of 1 or more	2	CSS Fragmentation 4 §3.3

Declarations in page-margin rules

Each page-margin at-rule contains declarations written as `property: value;`. Use properties that apply to page-margin boxes, including `content`, font and text properties, color, backgrounds, borders, margins, padding, dimensions and `vertical-align`. See the CSS Paged Media page-margin property list.

For example, set a running title in the top-center box and a page number in the bottom-center box:

```
@page {
    @top-center {
        content: "Annual report";
        font-weight: bold;
    }

    @bottom-center {
        content: "Page " counter(page);
        text-align: center;
    }
}
```

Links and generated content reference

The values list the syntax defined by the cited specifications.

Name	Description	Values	Default	Specification
`content`	Replaces an element's contents or supplies generated content for pseudo-elements and page margin boxes.	`normal`, none, an image that replaces the contents, a list of generated items, or `element()`; a list can include alternative text after `/`	`normal`	CSS Content 3 §1
`bookmark-label`	Sets the text of the bookmark of an element. It is also the default text of the entry of the element in a table of contents.	A list of generated content items	`content(text)`	CSS Content 3 §3.2

Name	Description	Values	Default	Specification
`bookmark-level`	Makes an element a bookmark of the PDF and an entry in a table of contents, and sets its level. Level 1 is the top level.	none or an integer of 1 or greater	`none`	CSS Content 3 §3.1
`bookmark-state`	Sets whether the bookmark of a block-level element shows its child bookmarks or hides them when the PDF opens.	`open` or `closed`	`open`	CSS Content 3 §3.3
`quotes`	Selects the quotation marks used by generated `open-quote` and `close-quote` content. With `auto`, the marks depend on the language of the text.	`auto`, `none`, `match-parent`, or one or more pairs of quoted strings	`auto`	CSS Content 3 §2.4.1
`string-set`	Assigns named values from an element for retrieval in page content.	none or comma-separated pairs of a name and one or more strings		CSS Content 3 §2.7.1
`string()`	Gives the value of a named string saved with `string-set`. Use it in the `content` property of a page-margin box.	A name, and an optional mode: `first`, `start`, `last` or `first-except`	`first`	CSS GCPM 3 §1.1.2
`content()`	Gives text of an element. Use it in the value of `string-set`.	`text`, `before`, `after` or `first-letter`	`text`	CSS GCPM 3 §1.1.1.1

Name	Description	Values	Default	Specification
`-cssmedia-link-counter()`	Gives the value of a counter at the destination of the link of the element, such as the number of a footnote that the link points to. CSSMedia extension.	The name of a counter, and an optional counter style.		CSSMedia extension
`-cssmedia-link`	Makes an XML element a link to another part of the document or to an external address. The values are joined in order to make the destination.	none, or one or more strings or `url()` values	none	CSSMedia extension
`link-parameters`	Supplies parameters for links to external resources.	none or a comma-separated list of `param()` values		CSS Link Parameters 1 §2.1
`leader()`	Fills the space between the content before it and the content after it on a line with a repeated pattern. Use it in `content`.	`dotted`, `solid`, `space` or a string		CSS Content 3 §2.5.1
`target-counter()`	Gives the value of a counter at the destination of a link, such as the page number of a heading. Use it in `content`.	A link, as `attr()`, a string or `url()`; a counter name; and an optional counter style		CSS Content 3 §2.6.1

Name	Description	Values	Default	Specification
`-cssmedia-toc()`	Generates a table of contents with an entry for each element that has a `bookmark-level`. Use it as the value of `content`. CSSMedia extension.	No arguments.		CSSMedia extension
`-cssmedia-toc-title`	Sets the title of the table of contents. Set it where `content` is `-cssmedia-toc()`. The title is also a bookmark if `@-cssmedia-toc-title` sets a `bookmark-level`. CSSMedia extension.	none or a list of generated content items	none	CSSMedia extension
`-cssmedia-toc-depth`	Sets the deepest level of `bookmark-level` that gets an entry in the table of contents. Set it where `content` is `-cssmedia-toc()`. CSSMedia extension.	`all` or an integer of 1 or greater	`all`	CSSMedia extension
`-cssmedia-toc-label`	Sets the text of the entry of an element in the table of contents. Set it on the element that has the `bookmark-level`. CSSMedia extension.	`auto` or a list of generated content items	`auto`, which uses the value of `bookmark-label`	CSSMedia extension

Name	Description	Values	Default	Specification
`-cssmedia-toc-divider`	Adds a line of text before the entry of an element in the table of contents, to group entries. Set it on the element that has the `bookmark-level`. CSSMedia extension.	none or a list of generated content items	none	CSSMedia extension
`-cssmedia-toc-leader`	Sets the leader between the text of an entry and its page number. Set it in `@-cssmedia-toc-entry`. CSSMedia extension.	`dotted`, `solid`, `space` or a string	`dotted`	CSSMedia extension
`@-cssmedia-toc-title`	Styles the title of the table of contents. CSSMedia extension.	A block of declarations.		CSSMedia extension
`@-cssmedia-toc-entry`	Styles the entries of the table of contents. Write a level after the name to style only the entries of that level. CSSMedia extension.	An optional integer of 1 or greater, and a block of declarations.		CSSMedia extension
`@-cssmedia-toc-divider`	Styles the dividers of the table of contents. Write a level after the name to style only the dividers of the entries of that level. CSSMedia extension.	An optional integer of 1 or greater, and a block of declarations.		CSSMedia extension

- A <`content-list`> is one or more generated content items, such as strings, quote keywords, counters, images, attribute values, or `content()` values. See CSS Content 3 and Named strings.

- A `<content-replacement>` is one image that replaces the element's normal contents. See CSS Content 3.
- The image form of `element()` uses an element ID to produce an image; the running-element form selects a named element for use in page margin boxes. See CSS Images 4 and CSS GCPM 3.
- A positive `<integer>` for `bookmark-level` is a whole number of at least 1. The top level is 1.
- A `<string>` is text between quotes. `url()` and `src()` provide URL values.
- A `<custom-ident>` is an author-chosen identifier, not a CSS-wide keyword or none.
- `param()` takes a dashed parameter name beginning with `--` and an optional `<declaration-value>`. If omitted, the parameter value is empty. A `<declaration-value>` is the sequence of CSS component values after a declaration's colon, excluding top-level semicolons and unmatched closing braces. See CSS Link Parameters 1 and CSS Syntax 3.

For examples of destinations, named strings, and generated content, see Links and Named strings.

Counters, list markers, and footnotes

Name	Description	Values	Default	Specification
`counter-reset`	Creates a new counter instance and initializes it.	none or one or more counter names, each optionally followed by an integer; the default value for each name is `0`		CSS Lists 3 §4.1
`counter-increment`	Increases each named counter, creating it if needed.	none or one or more counter names, each optionally followed by an integer; the default increment is `1`		CSS Lists 3 §4.2
`counter-set`	Sets the value of existing counters, creating a counter if it does not exist.	none or one or more counter names, each optionally followed by an integer; the default value is `0`		CSS Lists 3 §4.3
`list-style`	Sets the marker's position, image, and type.	Any order of `inside` or `outside`, an image or none, and a counter style, string, or none; any component can be omitted		CSS Lists 3 §3.6
`list-style-image`	Uses an image as the list marker.	An image or none		CSS Lists 3 §3.1

Name	Description	Values	Default	Specification
`list-style-position`	Places the marker inside the list item's contents or outside its principal box.	`inside` or `outside`		CSS Lists 3 §3.5
`list-style-type`	Chooses the marker's counter style or a literal marker string.	A counter-style name, a string, or none		CSS Lists 3 §3.4
`marker-side`	Chooses whether an outside marker's position follows the list item's directionality or its parent's.	`match-self` or `match-parent`		CSS Lists 3 §3.7
`@counter-style`	Defines a named counter style for markers and counter functions.	A name followed by counter-style descriptors		CSS Counter Styles 3 §3
`system` descriptor	Selects the algorithm that converts counter values to symbols.	`cyclic`, `numeric`, `alphabetic`, `symbolic`, `additive`, `fixed` with an optional starting integer, or `extends` with another counter-style name		CSS Counter Styles 3 §3.1
`symbols` descriptor	Lists the symbols used by the cyclic, fixed, symbolic, alphabetic, or numeric systems.	One or more symbols		CSS Counter Styles 3 §3.2
`additive-symbols` descriptor	Maps descending counter weights to symbols for the additive system.	Comma-separated pairs of a non-negative integer and a symbol, in descending weight order		CSS Counter Styles 3 §3.3
`negative` descriptor	Adds symbols before and optionally after a negative counter representation.	One or two symbols		CSS Counter Styles 3 §3.4
`prefix` descriptor	Adds a symbol before each marker.	One symbol		CSS Counter Styles 3 §3.7

Name	Description	Values	Default	Specification
`suffix` descriptor	Adds a symbol after each marker.	One symbol		CSS Counter Styles 3 §3.7
`range` descriptor	Limits the values that the counter style can represent before fallback is used.	`auto` or comma-separated pairs of integer bounds or `infinite`		CSS Counter Styles 3 §3.5
`pad` descriptor	Pads short representations to a minimum length with a chosen symbol.	A non-negative integer and a symbol		CSS Counter Styles 3 §3.6
`fallback` descriptor	Names the counter style used when this style cannot represent a value.	A counter-style name		CSS Counter Styles 3 §3.8
`footnote` counter	The counter that numbers the footnotes. Every footnote adds 1 to it. Change it with `counter-reset`, `counter-set` and `counter-increment`, in an element rule to number by section or in an @page rule to number by page.	Show it with `counter(footnote)`, with an optional counter style.	0 at the start of the document.	CSS GCPM 3 §2.5
`footnote-display`	Sets how a footnote body appears in the footnote area.	`block`, `inline`, or `compact`	`block`	CSS GCPM 3 §2.3
`footnote-policy`	Sets what happens when a footnote does not fit on the page of its call. `auto` continues it on the next pages, `line` moves the line with the call to the next page, and `block` moves the paragraph.	`auto`, `line`, or `block`	`auto`	CSS GCPM 3 §2.8

- A <counter-name> is an identifier naming a CSS counter. none and CSS-wide keywords are not counter names. See CSS Lists 3.
- An <integer> is a whole number without a fractional part. In the range descriptor, infinite means there is no bound in that direction. See CSS Values 4.
- An <image> is a URL, image function, or gradient that can provide image data. See CSS Images 3.
- A <counter-style> is a predefined counter style name, a name defined by @counter-style, or a symbols() function. See CSS Counter Styles 3.
- A <counter-style-name> is the identifier naming a predefined or author-defined counter style. See CSS Counter Styles 3.
- A <symbol> is a string, an identifier whose characters form the symbol, or an image. See CSS Counter Styles 3.

For example, this rule defines a two-symbol cycle. Replace item with an element name in your XML. The display declaration makes those elements list items, and their markers use the defined style.

```
@counter-style stars {
    system: cyclic;
    symbols: "★" "□";
    suffix: " ";
}

item {
    display: list-item;
    list-style-type: stars;
}
```

CSSMedia applies counter operations and prepares text marker content while building the box tree.

Named strings

This reference lists the property and the functions that CSSMedia uses for named strings.

Name	Description	Values	Default	Specification
string-set	Assigns named values from an element for retrieval in page content.	none or comma-separated pairs of a name and one or more strings		CSS Content 3 §2.7.1
content()	Gives text of an element. Use it in the value of string-set.	text, before, after or first-letter	text	CSS GCPM 3 §1.1.1.1

Name	Description	Values	Default	Specification
`string()`	Gives the value of a named string saved with `string-set`. Use it in the `content` property of a page-margin box.	A name, and an optional mode: `first`, `start`, `last` or `first-except`	`first`	CSS GCPM 3 §1.1.2

Links

This reference lists the property, the function and the pseudo-classes that CSSMedia uses for links.

Property and function

Name	Description	Values	Default	Specification
`-cssmedia-link`	Makes an XML element a link to another part of the document or to an external address. The values are joined in order to make the destination.	none, or one or more strings or `url()` values	none	CSSMedia extension
`-cssmedia-link-counter()`	Gives the value of a counter at the destination of the link of the element, such as the number of a footnote that the link points to. CSSMedia extension.	The name of a counter, and an optional counter style.		CSSMedia extension

Pseudo-classes

Name	Description	Values	In a PDF	Specification
`:link`, `:any-link`	Elements that are hyperlinks, because `-cssmedia-link` gives them a destination.		Matches	Selectors 4 §8.1, §8.2
`:local-link`	Hyperlinks that point to the same document. The optional number limits how many path segments are compared.	An optional non-negative integer in parentheses.	Matches	Selectors 5 §2.1
`:visited`, `:target`	Links that were visited, and the element that a link points to. A PDF has no browsing history and no current target.		Never matches	Selectors 4 §8.2, §8.3

Footnotes

This reference lists the properties, the rule, the counter and the pseudo-elements that CSSMedia uses for footnotes.

Properties, rule and counter

Name	Description	Values	Default	Specification
`float`	Takes a box out of the normal flow and places it at the side, at the top or at the bottom of its container. The value `footnote` makes the element a footnote.	`none`, `left`, `right`, `top`, `bottom`, `block-start`, `block-end`, `inline-start`, `inline-end`, `snap-block`, `snap-inline` or `footnote`	`none`	CSS Page Floats 3 §3.2
`footnote-display`	Sets how a footnote body appears in the footnote area.	`block`, `inline`, or `compact`	`block`	CSS GCPM 3 §2.3
`footnote-policy`	Sets what happens when a footnote does not fit on the page of its call. `auto` continues it on the next pages, `line` moves the line with the call to the next page, and `block` moves the paragraph.	`auto`, `line`, or `block`	`auto`	CSS GCPM 3 §2.8
`footnote` counter	The counter that numbers the footnotes. Every footnote adds 1 to it. Change it with `counter-reset`, `counter-set` and `counter-increment`, in an element rule to number by section or in an @page rule to number by page.	Show it with `counter(footnote)`, with an optional counter style.	0 at the start of the document.	CSS GCPM 3 §2.5

Name	Description	Values	Default	Specification
`@footnote`	Styles the footnote area at the bottom of the page. It goes inside an `@page` rule.	A block of declarations, such as borders, margins, padding and `max-height`.		CSS GCPM 3 §2.4

Pseudo-elements

Name	Description	Values	In a PDF	Specification
`::footnote-call`	The reference that replaces a footnote in the text. By default it is the value of the `footnote` counter, as a superscript.		Matches	CSS GCPM 3 §2.6
`::footnote-marker`	The marker at the start of a footnote body. By default it is the value of the `footnote` counter, followed by a period and a space.		Matches	CSS GCPM 3 §2.7

Table of contents

This reference lists the properties, functions and at-rules that CSSMedia uses for bookmarks and tables of contents.

Properties and functions

Name	Description	Values	Default	Specification
`bookmark-level`	Makes an element a bookmark of the PDF and an entry in a table of contents, and sets its level. Level 1 is the top level.	none or an integer of 1 or greater	none	CSS Content 3 §3.1
`bookmark-label`	Sets the text of the bookmark of an element. It is also the default text of the entry of the element in a table of contents.	A list of generated content items	`content(text)`	CSS Content 3 §3.2

Name	Description	Values	Default	Specification
`bookmark-state`	Sets whether the bookmark of a block-level element shows its child bookmarks or hides them when the PDF opens.	`open` or `closed`	`open`	CSS Content 3 §3.3
`-cssmedia-toc()`	Generates a table of contents with an entry for each element that has a `bookmark-level`. Use it as the value of `content`. CSSMedia extension.	No arguments.		CSSMedia extension
`-cssmedia-toc-title`	Sets the title of the table of contents. Set it where `content` is `-cssmedia-toc()`. The title is also a bookmark if `@-cssmedia-toc-title` sets a `bookmark-level`. CSSMedia extension.	none or a list of generated content items	none	CSSMedia extension
`-cssmedia-toc-depth`	Sets the deepest level of `bookmark-level` that gets an entry in the table of contents. Set it where `content` is `-cssmedia-toc()`. CSSMedia extension.	`all` or an integer of 1 or greater	`all`	CSSMedia extension
`-cssmedia-toc-label`	Sets the text of the entry of an element in the table of contents. Set it on the element that has the `bookmark-level`. CSSMedia extension.	`auto` or a list of generated content items	`auto`, which uses the value of `bookmark-label`	CSSMedia extension

Name	Description	Values	Default	Specification
`-cssmedia-toc-divider`	Adds a line of text before the entry of an element in the table of contents, to group entries. Set it on the element that has the `bookmark-level`. CSSMedia extension.	none or a list of generated content items	none	CSSMedia extension
`-cssmedia-toc-leader`	Sets the leader between the text of an entry and its page number. Set it in `@-cssmedia-toc-entry`. CSSMedia extension.	`dotted`, `solid`, space or a string	`dotted`	CSSMedia extension
`leader()`	Fills the space between the content before it and the content after it on a line with a repeated pattern. Use it in `content`.	`dotted`, `solid`, space or a string		CSS Content 3 §2.5.1
`target-counter()`	Gives the value of a counter at the destination of a link, such as the page number of a heading. Use it in `content`.	A link, as `attr()`, a string or `url()`; a counter name; and an optional counter style		CSS Content 3 §2.6.1

At-rules

Name	Description	Values	Default	Specification
`@-cssmedia-toc-title`	Styles the title of the table of contents. CSSMedia extension.	A block of declarations.		CSSMedia extension
`@-cssmedia-toc-entry`	Styles the entries of the table of contents. Write a level after the name to style only the entries of that level. CSSMedia extension.	An optional integer of 1 or greater, and a block of declarations.		CSSMedia extension

Name	Description	Values	Default	Specification
`@-cssmedia-toc-divider`	Styles the dividers of the table of contents. Write a level after the name to style only the dividers of the entries of that level. CSSMedia extension.	An optional integer of 1 or greater, and a block of declarations.		CSSMedia extension

Images

This reference lists the properties that CSSMedia uses to size, fit and draw images.

Image properties

Name	Description	Values	Default	Specification
`image-resolution`	Sets how many pixels of a raster image make one CSS pixel, which sets its natural size. `from-image` uses the resolution that the file states.	`from-image`, a resolution such as `2dppx` or `192dpi`, or both	`1dppx`	CSS Images 4 §6.1
`object-fit`	Chooses how an image fills its box: stretched, scaled to fit, scaled to cover, or at its natural size.	`fill`, `none`, `contain`, `cover` or `scale-down`; `scale-down` can follow `contain` or `cover`	`fill`	CSS Images 4 §5.1
`object-position`	Sets where an image goes in its box when it does not fill it.	One to four values: keywords, lengths and percentages, as in `background-position`	`50% 50%`	CSS Images 3 §4.6
`image-orientation`	Chooses whether an image is turned as its file says, and can turn or mirror it.	`from-image`, `none`, or an angle and `flip`	`from-image`	CSS Images 3 §5.1
`image-rendering`	Chooses how the pixels of an image are drawn when it is shown larger than its size: smoothed, with sharp edges, or enlarged by whole multiples of its pixels.	`auto`, `smooth`, `high-quality`, `pixelated` or `crisp-edges`	`auto`	CSS Images 3 §5.2

Name	Description	Values	Default	Specification
`image-animation`	Chooses whether an animated image plays. A PDF does not move, so it has no effect.	`normal`, `paused`, `stopped` or `running`	`normal`	CSS Image Animation 1

SVG

This reference lists the elements of SVG that CSSMedia draws and the properties that style them.

SVG elements

Element	Description	Attributes
`svg`	A drawing, or a nested drawing with its own size and scale.	`xmlns`, `x`, `y`, `width`, `height`, `viewBox`, `preserveAspectRatio`
`g`	Groups elements, so that they share styles and a transform.	`transform`
`defs`	Holds elements that are defined to be used later and are not drawn.	None
`use`	Draws a copy of another element.	`href`, `x`, `y`, `width`, `height`
`symbol`	A shape with its own `viewBox`, drawn by a `use` element.	`viewBox`, `preserveAspectRatio`, `refX`, `refY`
`switch`	Draws its first child that has no conditional attributes.	`systemLanguage`, `requiredExtensions` on the children
`rect`	A rectangle, with optional rounded corners.	`x`, `y`, `width`, `height`, `rx`, `ry`
`circle`	A circle.	`cx`, `cy`, `r`
`ellipse`	An ellipse.	`cx`, `cy`, `rx`, `ry`
`line`	A straight line.	`x1`, `y1`, `x2`, `y2`
`polyline`	An open shape made of straight lines.	`points`
`polygon`	A closed shape made of straight lines.	`points`

Element	Description	Attributes
`path`	A shape made of lines, curves and arcs.	`d`, `pathLength`
`linearGradient`	A paint that changes color along a line.	`x1`, `y1`, `x2`, `y2`, `gradientUnits`, `gradientTransform`, `spreadMethod`, `href`
`radialGradient`	A paint that changes color from a center to a circle.	`cx`, `cy`, `r`, `fx`, `fy`, `gradientUnits`, `gradientTransform`, `spreadMethod`, `href`
`stop`	A color of a gradient, at a position.	`offset`
`pattern`	A paint that repeats a small drawing.	`x`, `y`, `width`, `height`, `patternUnits`, `patternContentUnits`, `patternTransform`, `viewBox`, `preserveAspectRatio`, `href`
`marker`	A mark that is drawn on the vertices of a shape.	`markerWidth`, `markerHeight`, `refX`, `refY`, `orient`, `markerUnits`, `viewBox`, `preserveAspectRatio`
`clipPath`	An area that clips other elements.	`clipPathUnits`
`mask`	A drawing whose light or opaque parts show the masked element.	`x`, `y`, `width`, `height`, `maskUnits`, `maskContentUnits`
`text`	Text, in a line, wrapped to a length or inside a shape.	`x`, `y`, `dx`, `dy`, `rotate`, `textLength`, `lengthAdjust`
`tspan`	A part of a text with its own style or position.	`x`, `y`, `dx`, `dy`, `rotate`, `textLength`, `lengthAdjust`
`textPath`	Text that follows the outline of a shape.	`href`, `startOffset`, `side`
`image`	A PNG, JPEG, WebP or SVG file.	`href`, `x`, `y`, `width`, `height`, `preserveAspectRatio`
`a`	A link around the elements or the text it holds.	`href`
`style`	A style sheet for the drawing.	`type`, `media`

Geometry properties

Name	Description	Values	Default	Specification
x, y	Set the position of a `rect`, `image`, `svg`, `symbol` or `use` element.	A length or a percentage	`0`	SVG 2
cx, cy	Set the center of a `circle` or an `ellipse`.	A length or a percentage	`0`	SVG 2
r	Sets the radius of a `circle`.	A length or a percentage	`0`	SVG 2
rx, ry	Set the radii of an `ellipse`, or the rounded corners of a `rect`. With `auto` the radius takes the value of the other one.	A length, a percentage or `auto`	`auto`	SVG 2
d	Sets the data of a `path`.	none or a string of path data	none	SVG 2
`path-length`	Sets the length that a shape counts as having, for the lengths of its dashes.	none or a number	none	SVG 2

Paint properties

Name	Description	Values	Default	Specification
`fill`	Sets the paint of the inside of a shape or of the characters of a text.	A color, none, `currentColor` or `url(#id)`, optionally followed by a color that is used when the reference names nothing	`black`	SVG 2
`fill-opacity`	Sets the opacity of the fill.	A number from 0 to 1, or a percentage	`1`	SVG 2
`fill-rule`	Chooses which areas of a shape are inside it.	`nonzero` or evenodd	`nonzero`	SVG 2
`stroke`	Sets the paint of the outline of a shape or of a text.	As `fill`	none	SVG 2
`stroke-width`	Sets the thickness of the outline.	A length, a percentage or a number	`1`	SVG 2
`stroke-opacity`	Sets the opacity of the outline.	A number from 0 to 1, or a percentage	`1`	SVG 2

Name	Description	Values	Default	Specification
`stroke-linecap`	Sets the shape of the ends of an open line.	`butt`, `round` or `square`	`butt`	SVG 2
`stroke-linejoin`	Sets the shape of the corners of the outline.	`miter`, `round` or `bevel`	`miter`	SVG 2
`stroke-miterlimit`	Sets the length at which a sharp corner turns into a bevel, as a multiple of the thickness.	A number of 1 or more	`4`	SVG 2
`stroke-dasharray`	Makes the outline dashed, with a list of lengths for dashes and gaps.	none or a list of lengths, percentages or numbers	`none`	SVG 2
`stroke-dashoffset`	Moves the start of the dash pattern along the outline.	A length, a percentage or a number	`0`	SVG 2
`paint-order`	Chooses the order in which the fill, the stroke and the markers are painted.	`normal`, or `fill`, `stroke` and `markers` in the order you want	`normal`	SVG 2
`vector-effect`	With `non-scaling-stroke` the outline keeps its thickness when a transform scales the shape.	none or `non-scaling-stroke`	`none`	SVG 2
`color-interpolation`	Chooses the color space in which the colors of a gradient mix.	`auto`, `sRGB` or `linearRGB`	`sRGB`	SVG 2
`stop-color`	Sets the color of a `stop` of a gradient.	A color	`black`	SVG 2
`stop-opacity`	Sets the opacity of a `stop` of a gradient.	A number from 0 to 1, or a percentage	`1`	SVG 2

Name	Description	Values	Default	Specification
`marker`	Sets `marker-start`, `marker-mid` and `marker-end` at once.	none or `url(#id)`	none	SVG 2
`marker-start`, `marker-mid`, `marker-end`	Name the marker that is drawn on the first vertex, on the vertices between, and on the last vertex of a shape.	none or `url(#id)`	none	SVG 2

Text properties of a drawing

Name	Description	Values	Default	Specification
`text-anchor`	Chooses which part of a line of text is at its x position.	`start`, `middle` or `end`	`start`	SVG 2
`dominant-baseline`	Chooses the baseline of the text that is placed at its y position.	`auto`, `alphabetic`, `middle`, `central`, `hanging`, `ideographic`, `mathematical`, `text-top` or `text-bottom`	`auto`	SVG 2
`baseline-shift`	Raises or lowers a part of a line of text.	`super`, `sub`, a length or a percentage	`0`	SVG 2
`inline-size`	On a `text` element, sets the length of its lines, and the text wraps to it.	`auto`, a length or a percentage	`auto`	SVG 2
`shape-inside`	On a `text` element, pours the text into a shape.	`auto`, a basic shape, `url(#id)` of a shape, or the URL of an image	`auto`	CSS Shapes 2

MathML

This reference lists the properties, the elements and the attributes that CSSMedia uses to lay out and draw MathML formulas.

Math properties and values

Name	Description	Values	Default	Specification
`math-depth`	The script level of an element: how many steps of script size it is below the text. It is inherited. The attribute `scriptlevel` sets it.	`auto-add`, `add(an integer)` or an integer	`0`	MathML Core §4.5
`math-shift`	Chooses how high the superscripts of an element are raised. `compact` raises them less, to keep them clear of what is above.	`normal` or `compact`	`normal`	MathML Core §4.4
`math-style`	Chooses display style or compact style. `normal` gives fractions, limits and big operators their full size and room. The attribute `displaystyle` sets it.	`normal` or `compact`	`normal`	MathML Core §4.3
`display`	The values `block math` and `inline math` make an element a formula that is a block or inline. On an element that is not a MathML element, they compute to `block flow` and `inline flow`.	`block math` or `inline math`	`inline math` on `math`, `block math` on other MathML elements	MathML Core §4.1
`font-size`	The value `math` gives the size that the script level of the element calls for.	`math`	`math` on MathML elements	MathML Core §4.5
`text-transform`	The value `math-auto` turns a single letter into its mathematical italic form.	`math-auto`	`math-auto` on `mi`	MathML Core §4.2

MathML elements

Element	Description	Attributes
`math`	A formula. It is a block or inline. It accepts the attributes of `mstyle`.	`xmlns`, `display`, `dir`, `maxwidth`, `overflow`, `indentalignfirst`, `indentshiftfirst`
`mi`	An identifier. One letter is drawn in italic.	`mathvariant`

Element	Description	Attributes
`mn`	A number.	`mathvariant`
`mo`	An operator, a fence, a separator or an accent. It has space around it, and it stretches.	`form`, `stretchy`, `symmetric`, `largeop`, `movablelimits`, `accent`, `lspace`, `rspace`, `minsize`, `maxsize`, `mathvariant`, `linebreak`, `linebreakstyle`, `linebreakmultchar`, `lineleading`, `indentalign`, `indentshift`, `indenttarget`, `indentalignfirst`, `indentshiftfirst`, `indentalignlast`, `indentshiftlast`
`mtext`	Text that is not math.	`mathvariant`
`ms`	A string, shown between quotes with the quotes inside it escaped.	`lquote`, `rquote`, `mathvariant`
`mspace`	An empty space.	`width`, `height`, `depth`, `linebreak`, `indentalign`, `indentshift`, `indenttarget`, `indentalignfirst`, `indentshiftfirst`, `indentalignlast`, `indentshiftlast`
`mglyph`	An image, in a token or in a row.	`src`, `alt`, `width`, `height`, `valign`
`mrow`	A group of elements in a row. An element that is not in this table is laid out as an `mrow`.	None
`mstyle`	A group that sets styles for what it holds.	`scriptlevel`, `displaystyle`, `scriptminsize`, `scriptsizemultiplier`, and the style attributes of the other elements
`mpadded`	Changes the size of the box of its content.	`width`, `height`, `depth`, `lspace`, `voffset`
`mphantom`	Takes the room of its content and does not show it.	None
`merror`	Shows its content in a box with a red border and a light yellow background.	None

Element	Description	Attributes
`mfrac`	A fraction.	`linethickness`, `numalign`, `denomalign`, `bevelled`
`msqrt`, `mroot`	A square root, and a root with an index.	None
`msub`, `msup`, `msubsup`	A base with a subscript, a superscript, or both.	`subscriptshift`, `superscriptshift`
`mmultiscripts`, `mprescripts`	A base with pairs of scripts after it and, after `mprescripts`, before it.	`subscriptshift`, `superscriptshift`
`munder`, `mover`, `munderover`	A base with a part under it, over it, or both.	`accent`, `accentunder`, `align`
`mtable`	A table or a matrix, laid out as an inline table.	`columnalign`, `rowalign`, `columnspacing`, `rowspacing`, `columnlines`, `rowlines`, `frame`, `framespacing`, `width`, `columnwidth`, `equalrows`, `equalcolumns`, `align`
`mtr`	A row of a table.	`columnalign`, `rowalign`
`mtd`	A cell of a table.	`columnspan`, `rowspan`, `columnalign`, `rowalign`
`mfenced`	An expression between fences, with separators between its children.	`open`, `close`, `separators`
`menclose`	Draws lines, arrows or strikes around its content.	`notation`
`mstack`, `mlongdiv`, `msgroup`, `msrow`, `mscarries`, `mscarry`, `msline`	Stacks of numbers for elementary arithmetic, and long division.	`mstack` and `mlongdiv`: `stackalign`, `charalign`, `charspacing`, `align`, `decimalcharacter` (or `decimalpoint`), and `longdivstyle` on `mlongdiv`. `msgroup`: `position`, `shift`. `msrow`: `position`. `mscarries`: `position`, `location`, `crossout`, `scriptsizemultiplier`. `mscarry`: `location`, `crossout`. `msline`: `position`, `length`, `leftoverhang`, `rightoverhang`, `mslinethickness`

Element	Description	Attributes
`semantics`, `annotation`, `annotation-xml`	A formula with other representations. Only the first child of `semantics` is drawn.	`encoding`
`maction`	Shows one of its children.	`actiontype`, `selection`
`a`	A link. Any MathML element with an `href` is a link.	`href`

MathML attributes of any element

Attribute	Description	Values
`class`	A list of class names for the selectors of the style sheet.	Names separated by spaces
`style`	Declarations for the element. They win over the rules of the style sheet.	A list of CSS declarations
`mathcolor`	The color of the text.	A color
`mathbackground`	The color behind the element.	A color
`mathsize`	The size of the text.	A length, a percentage, `small`, `normal` or `big`
`mathvariant`	The style of the letters of a token.	`normal`, `italic`, `bold`, `bold-italic`, `double-struck`, `script`, `bold-script`, `fraktur`, `bold-fraktur`, `sans-serif`, `bold-sans-`

Attribute	Description	Values
		`serif`, `sans-serif-italic`, `sans-serif-bold-italic`, `monospace`, `initial`, `tailed`, `looped`, `stretched`
`displaystyle`	Display style or compact style. It sets `math-style`.	`true` or `false`
`scriptlevel`	The script level. It sets `math-depth`.	An unsigned integer, or an integer with a sign that is added to the level of the parent
`dir`	The direction of the text.	`ltr` or `rtl`
`href`	The target of a link.	A URL
`scriptminsize`	The least size of the text of a script, 8pt by default.	A length
`scriptsizemultiplier`	The factor from the size of a level of script to the size of the next one.	A number
`infixlinebreakstyle`	Where an operator goes when a line breaks at it, for the operators that have `linebreakstyle="infixlinebreakstyle"`.	`before`, `after` or `duplicate`
`indentalign`, `indentshift`, `indenttarget`, `indentalignfirst`, `indentshiftfirst`, `indentalignlast`, `indentshiftlast`, `lineleading`	The indentation and the spacing of the lines of a broken formula. On `mstyle` and `math` they are the defaults of the breaks inside.	As on `mo`
`decimalcharacter` (and `decimalpoint`)	The character that is the decimal point of a stack.	One character

Content MathML elements

Element	Description	Attributes
cn, sep	A number, and the separator of its parts.	type, base
ci, csymbol	A variable and a symbol. A csymbol that names a symbol of a content dictionary is that operator.	cd, definitionURL
cs	A string, shown between quotes.	lquote, rquote
apply, bind	An operator applied to arguments, and a binder applied to bound variables and an expression.	None
bvar, lowlimit, uplimit, interval, condition, domainofapplication, degree, momentabout, logbase	A bound variable, and the qualifiers of an operator.	closure on interval
set, list, vector, matrix, matrixrow, piecewise, piece, otherwise, lambda	Collections, matrices, piecewise functions and lambda expressions.	None
share, cerror, cbytes	A reference to another element of the formula, an error, and bytes, which are not drawn.	src on share
The empty operator, constant and set elements	The operators, the constants and the number sets of Content MathML, which the next table lists with their notation.	type on tendsto

Content MathML operators, constants and sets

Elements	Notation	Specification
plus, minus, divide, rem, compose	+, −, /, mod and □ between the arguments. minus with one argument is a sign. A sum that is an operand of a product or of a negation is in parentheses.	MathML 4 §4.3.6.1

Elements	Notation	Specification
`times`, `power`, `root`, `exp`, `factorial`, `quotient`	The arguments next to each other; a superscript; a radical, with the index of a `degree` qualifier; e with a superscript; ! after the argument; □a/b□.	MathML 4 §4.3.5.1
`and`, `or`, `implies`, `xor`, `not`	∧, ∨ and ⇒ between the arguments, `xor` as a function, and ¬ before the argument.	MathML 4 §4.3.6.2
`eq`, `neq`, `lt`, `gt`, `leq`, `geq`, `approx`, `equivalent`, `factorof`, `tendsto`	=, ≠, <, >, ≤, ≥, □, ≡, | and → between the arguments. `tendsto` with `type="above"` raises a + over its last argument.	MathML 4 §4.3.6.3
`union`, `intersect`, `setdiff`, `cartesianproduct`, `subset`, `prsubset`, `notsubset`, `notprsubset`, `in`, `notin`	∪, ∩, □, ×, ⊆, ⊂, □, ⊄, ∈ and ∉ between the arguments.	MathML 4 §4.3.6.5
`vectorproduct`, `scalarproduct`, `outerproduct`, `transpose`, `inverse`, `conjugate`, `determinant`	×, a dot and ⊗ between the arguments; a superscript T; a superscript (−1); a bar over the argument; det with its argument.	MathML 4 §4.3.6.4
`abs`, `card`, `floor`, `ceiling`, `mean`	The argument between |...|, |...|, □...□, □...□ and □...□.	MathML 4 §4.3.7.2
`sin`, `cos`, `tan`, `sec`, `csc`, `cot`, `sinh`, `cosh`, `tanh`, `sech`, `csch`, `coth`, `arcsin`, `arccos`, `arctan`, `arcsec`, `arccsc`, `arccot`, `arcsinh`, `arccosh`, `arctanh`, `arcsech`, `arccsch`, `arccoth`, `ln`, `log`	The name of the function and its argument, without parentheses when the argument is a single token. `log` takes the base of a `logbase` qualifier as a subscript.	MathML 4 §4.3.7.6
`gcd`, `lcm`, `median`, `mode`, `arg`, `domain`, `codomain`, `image`, `selector`	The name and its arguments in parentheses. `selector` is the first argument with the others as a subscript.	MathML 4 §4.3.5.4
`min`, `max`, `variance`, `sdev`, `real`, `imaginary`, `moment`	The name and the arguments in braces, or a body and a condition; σ^2 and σ; ℜ and ℑ; □x^n□ with the `momentabout` point under it.	MathML 4 §4.3.5.12
`sum`, `product`, `and`, `union`, `intersect` with a `bvar` or a qualifier	∑, ∏, □, □ and □ with the limits, the condition or the domain under and over the sign, then the body.	MathML 4 §4.3.5

Elements	Notation	Specification
`int`, `diff`, `partialdiff`, `limit`	An integral with its limits, d and the variable after the body; f′ or d(body)/dx and its higher orders; *∂* with the order of each variable; lim with the variable and its limit under it.	MathML 4 §4.3.8.1
`divergence`, `grad`, `curl`, `laplacian`	div, grad, curl and ∇^2 applied to the argument. With a `bvar` the argument is a function of the variables.	MathML 4 §4.3.7.7
`forall`, `exists`, `lambda`	∀, ∃ and λ, the variables, a dot, and the expression in parentheses. A condition or a domain restricts the variables.	MathML 4 §4.3.10.1
`set`, `list`, `vector`, `matrix`, `matrixrow`, `interval`, `piecewise`, `piece`, `otherwise`	{...}, (...), a column in parentheses, a table in parentheses, the row of a matrix, an interval with the ends that `closure` chooses, and a brace with a table of values and conditions.	MathML 4 §4.3.5
`exponentiale`, `imaginaryi`, `notanumber`, `true`, `false`, `pi`, `eulergamma`, `infinity`, `emptyset`, `ident`	e, i, NaN, true, false, π, γ, ∞, ∅ and id.	MathML 4 §4.3.9.1
`integers`, `reals`, `rationals`, `naturalnumbers`, `complexes`, `primes`	Z, R, Q, N, C and P in double-struck letters.	MathML 4 §4.3.9.2
Any operator, constant or set above, written alone, without `apply`	The name of the operator.	MathML 4 §4.2

Glossary

Bookmark

A title in the navigation tree of a PDF. PDF viewers, such as Adobe Acrobat, show the bookmarks in a side panel. Selecting a title goes to its page. Bookmarks can have child bookmarks.

CSS

Cascading Style Sheets (CSS) is the language, defined by the W3C, that describes how elements of a document are presented. A stylesheet is a list of rules. Each rule has a selector that chooses elements and declarations that set their properties. CSSMedia reads a CSS file to decide how to format an XML document in a PDF.

DTD

A Document Type Definition (DTD) is a grammar, defined by the XML specification, that declares the elements and attributes an XML document may use. A document points to its DTD in its document type declaration. When the DTD is not at the location the document names, an OASIS Catalog tells CSSMedia where to find it.

Font face

One style of a font family, such as its bold italic style. A font file holds one face, or several faces in a font collection. CSS Fonts 4 defines how a stylesheet declares faces.

Glyph

The shape that a font draws for a character or for a group of characters. One character can have several glyphs, such as a plain letter and a small capital.

Kerning

The adjustment of the space between two letters, so that pairs such as "AV" do not look too far apart.

Leader

A row of dots, or of another repeated pattern, that fills the space between two parts of a line. A table of contents uses a leader to join the title of an entry to its page number. CSS Content 3 defines leaders.

Ligature

A single glyph that replaces two or more letters, such as the glyph for "fi". Fonts use ligatures to avoid collisions between letters.

MathML

Mathematical Markup Language (MathML) is a W3C XML language for writing mathematical notation. A MathML formula describes the structure of an expression, such as fractions, roots and subscripts. CSSMedia draws MathML found in a document as an image.

Namespace

An XML namespace is a URI that identifies the vocabulary an element or attribute belongs to. An XML document declares it with an `xmlns` attribute, so elements with the same name in different vocabularies do not collide.

OASIS Catalog

An OASIS Catalog is an XML file, defined by the OASIS XML Catalogs standard, that maps the identifiers used in documents to the locations where the files are stored. A catalog lets a parser find a DTD, an XML Schema or a RelaxNG schema without downloading it. Pass a catalog to CSSMedia with the `-g` option.

OpenType

The font format that most fonts use today. An OpenType font holds glyphs and tables of features, such as ligatures, alternate shapes and kerning. Each feature has a four-letter tag. The OpenType specification defines the format.

PDF

Portable Document Format (PDF) is the file format, standardized as ISO 32000, for documents that look the same on any device. A PDF holds pages with text, fonts, images and vector graphics at fixed positions, and may also hold links and bookmarks. CSSMedia writes a PDF file from an XML file and a CSS file.

RelaxNG

RelaxNG is a schema language for XML, standardized as ISO/IEC 19757-2. A RelaxNG schema describes the structure of a document in an XML syntax or in a compact non-XML syntax. It is an alternative to a DTD and to an XML Schema, and it can be located with an OASIS Catalog.

Variable font

A font whose design changes along axes, such as weight, width and slant. One variable font file can replace many ordinary font files. Each axis has a four-letter tag and a range of values. The OpenType specification describes variable fonts.

SVG

Scalable Vector Graphics (SVG) is a W3C format for two-dimensional vector graphics, written as XML. An SVG drawing describes shapes, paths, text and images that can be scaled without losing quality. CSSMedia draws SVG found in a document as an image.

XML

Extensible Markup Language (XML) is a W3C standard for writing structured documents. An XML document is a tree of elements, with attributes and text, whose names are chosen by the author or by a vocabulary such as DITA or DocBook. CSSMedia reads an XML file as the content of the PDF.

XML Schema

XML Schema is a W3C language for describing the structure of XML documents and the data types of their content. A schema is itself an XML file, and it is more expressive than a DTD. Like a DTD, a schema can be located with an OASIS Catalog.